

Segmentation in Pharmaceutical Markets: Evidence from Contraceptives*

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Abstract

Pharmaceutical markets are often divided between over-the-counter (OTC) and health insurance segments. This paper shows that segmentation of this kind dampens pharmaceutical demand by softening product competition and raising prices relative to a pooled market, an effect amplified by information frictions about drug quality. Our setting is the hormonal contraceptive market in Colombia, where a price ceiling policy generates variation that allows us to separately identify price sensitivity, preferences for quality, and information frictions. Using a structural model, we find that when product variety is held fixed, moving to an OTC-only market would raise contraceptive demand by 8% and reduce fertility by 0.7 births per woman. However, when manufacturers eliminate redundant products, moving to an OTC-only market would reduce demand by 2% and raise fertility by 0.2 births per woman. Effects are qualitatively similar when moving to an insurance-only market.

Keywords: Market segmentation, Contraception, Information frictions, Price ceilings.

JEL codes: I18, I12, L10.

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1 Introduction

Pharmaceutical markets are often divided between two very different segments. A drug may be covered by health insurance, sold over the counter (OTC), or available through both. Which of these segments emerges depends on two distinct regulatory decisions, insurance formularies and OTC approval, that are typically made by different authorities with little consideration of how they interact. As a result, even drugs within the same therapeutic class can be separated across market segments with very different prices, access conditions, and sets of competing products.

This paper shows that market segmentation of this kind softens product competition and allows pharmaceutical manufacturers to price drugs higher than in a pooled market. Segmentation thus offsets the access gains that OTC approval is meant to deliver and limits the ability of restricted insurance formularies to contain pharmaceutical spending.¹ Moreover, unlike markets where segmentation is common and consumers are well-aware of product attributes (Karle et al., 2020; Marshall, 2015), pharmaceutical markets feature widespread information frictions about product quality (Crawford and Shum, 2005), which further amplify the effects of segmentation.

We examine these effects in the hormonal contraceptive market where information frictions arise naturally due to the risks of experimentation.² This setting has direct policy relevance: contraceptives are available through both OTC and insurance segments in 56 countries (Grindlay et al., 2013), and the set of drug classes subject to this form of segmentation is actively expanding. For example, in December 2024, the U.S. Food and Drug Administration (FDA) finalized the Additional Conditions for Nonprescription Use (ACNU) rule, creating a pathway for drugs that previously required medical supervision to be approved for OTC sale (FDA, 2024). Under this rule, a progestin-only pill became available OTC for the first time, even as the Affordable Care Act already mandated comprehensive insurance coverage of contraceptives (FDA, 2024).

¹Many countries have expanded OTC availability for drug classes where unsupervised use is safe to alleviate health systems from consultations that are not always clinically necessary.

²Information frictions are common in contraceptive use because learning about contraceptive quality requires either experimenting with different products—at the risk of unintended pregnancy if a method fails—or a level of health literacy that is unevenly distributed across women (D’Souza et al., 2022; Colgan et al., 2015).

Our empirical setting is Colombia, which is uniquely informative for studying this question. In the OTC segment, women pay full retail prices and have access to all government-approved contraceptives. In the insurance segment, women pay nothing out of pocket but face a restricted formulary and hassle costs from interacting with the healthcare system.³ Our analysis combines product-level data on monthly retail prices and sales from IQVIA Colombia for the OTC segment with individual-level data on enrollment and health claims covering nearly half of the Colombian population for the insurance segment, both between 2018 and 2019.

We begin by documenting the consequences of market segmentation in reduced form, exploiting a 2019 policy in which the Colombian government imposed wholesale price ceilings on 42 contraceptives, half of which were marketed as higher-quality products because of their milder side effects. Using an event study design that compares targeted with untargeted products, we find that retail prices fell 45% for products offered exclusively OTC and by 59% for products simultaneously available through insurance. The difference in treatment effects reflects the fact that pre-regulation prices for OTC-only products were substantially above their ceilings, suggesting that segmentation allows manufacturers to set relatively high prices when the insurance segment does not offer a comparable alternative.

Replicating the event study exercise in the insurance segment data reveals that wholesale prices for targeted products fell 28% and their demand nearly doubled by the end of the sample period. However, changes in demand in this segment cannot be explained by price reductions as women face zero out-of-pocket prices.

We provide evidence that this demand response reflects mainly a reduction in information frictions. Using a survey of 296 reproductive-age women in Bogotá, we find that although 80.4% of respondents recognize at least one brand, 32.4% cannot correctly identify the route of administration of any brand, indicating that information frictions are widespread. Knowledge is higher among women who finance their household's health insurance (contributors) than among dependents, further indicating a correlation

³Transaction costs, time costs of scheduling doctor appointments, and travel costs to insurer medication-dispensing facilities are examples of hassle costs women face when claiming contraceptives through insurance.

between insurance status and information frictions.⁴ The claims data corroborate the survey: better-informed contributors exhibit 25% higher demand for targeted products (i.e., higher-quality products) than less-informed dependents, and this gap halves after the policy, suggesting that the regulation acted as an information signal that updated the beliefs of less-informed women.

We also find evidence of hassle costs in the insurance segment. Our survey shows that among women who obtain contraceptives through insurance, 81.8% report that obtaining the product takes more than one hour, compared with 81.2% of OTC purchasers who obtain it in under one hour.⁵ In the claims data, measures of hassle costs are also strongly correlated with the wholesale prices insurers pay manufacturers, consistent with insurers imposing greater administrative barriers for more expensive products, and suggesting that wholesale prices serve as a proxy for hassle costs in the structural model.

To quantify the equilibrium effects of segmentation, we develop and estimate a structural model of the contraceptive market. We model demand among women seeking to avoid pregnancy using a random-coefficients logit framework in which women choose among available hormonal contraceptives and an outside option to maximize their indirect utility. Demand depends on contraceptive prices, hassle costs, and beliefs about product quality. Wholesale prices proxy for hassle costs in the insurance segment. Contributor status, in turn, determines women's information about product quality. We assume women are Bayesian agents who update their beliefs about contraceptive quality upon observing whether a product is regulated. On the supply side, pharmaceutical manufacturers sell through both market segments and set segment-specific prices, competing Nash-Bertrand in the OTC segment and making take-it-or-leave-it offers to a representative insurer in the insurance segment.

Our estimates show that women place 84% weight on their prior beliefs about product quality and the remaining 16% on quality signals, implying that most women remain

⁴The Colombian health insurance system distinguishes between enrollees who finance coverage for themselves and their eligible family members through payroll deductions or direct payments (*contributors*) and the family members covered through them (dependents). The administrative claims analysis compares contributors and dependents because it is restricted to enrollees in the contributory insurance scheme.

⁵In Colombia, medications approved by insurers must be claimed at insurer-owned facilities instead of pharmacies as in the United States, and wait times at these facilities are relatively long (see [El Colombiano, Dec. 21, 2023](#)).

poorly informed in the absence of a policy shock. The median own-price elasticity in the OTC segment equals -1.77 and declines in absolute value with age. These estimates are consistent with those from the randomized controlled trial in [Bailey et al. \(2025\)](#), which studies how out-of-pocket cost subsidies affect contraceptive method choice among uninsured women in the U.S. Women also face substantial hassle costs when obtaining contraceptives through insurance. Each additional dollar in wholesale acquisition costs is associated with \$2.98 in hassle costs. On the supply side, we find that the average markup equals 62% of the retail price in the OTC segment and 13% of the wholesale price in the insurance segment.

Using the model estimates, we simulate counterfactual scenarios that eliminate segmentation by moving to an OTC-only or an insurance-only market. The OTC-only scenario can be interpreted as granting non-prescription status to all drug classes, while the insurance-only scenario corresponds to requiring the national insurance formulary to cover all products. Additionally, we remove information frictions to separate their contribution to welfare from that of retail prices and hassle costs.

We find that market segmentation dampens contraceptive demand relative to a pooled market where we hold product variety fixed. Moving to an OTC-only setting increases demand by 0.137 million women per year or 8% relative to baseline. These effects are driven by lower equilibrium prices that result from increased product competition. Moving to an insurance-only market yields qualitatively similar results. Total demand increases by roughly 0.278 million women per year, or 16% relative to baseline, as demand reallocates toward longer-acting methods, which become relatively cheaper. In both counterfactuals, the changes in demand and prices are more pronounced when information frictions are additionally removed, suggesting that manufacturers exploit these frictions to sustain higher markups in the segmented market.

The assumption that product variety is fixed after pooling the market is important for equilibrium outcomes. Allowing manufacturers to discontinue redundant products would instead reduce contraceptive demand in the pooled market and raise prices. Hence, maintaining product variety is key to achieve the benefits of pooling the market.

We conclude by examining the impacts on total fertility rates. We draw from [Ennis and](#)

[Serna \(2026\)](#), who estimate the causal effect of the price ceilings on fertility by comparing Colombian municipalities that are more or less exposed to the policy based on their baseline share of women who consume products later targeted by the policy. Using their estimates, we find that when product variety is held fixed, moving to an OTC-only market would reduce fertility by 0.713 births per woman. However, when product variety adjusts, fertility would increase by 0.168 births per woman in the OTC-only market and 0.219 in the insurance-only market.

Our paper makes three contributions to the literature. First, the literature on market segmentation has mostly studied markets characterized by adverse selection ([Shahidinejad et al., 2025](#)), such as health insurance, and found that pooling the market results in lower premiums ([Fleitas et al., 2019](#); [Dickstein et al., 2024](#)). We contribute by showing that the effects of pooling are similar in markets without selection but where segments differ in prices and hassle costs.

Second, the industrial organization literature studying information frictions has mostly focused on its impact on product demand ([Brown et al., 2023](#); [Dickstein et al., 2023](#); [Brown and Jeon, 2024](#); [Gupta and Cao, 2025](#)). Within pharmaceutical markets, the literature has mainly studied information frictions as an explanation for why consumers buy branded drugs over generics ([Handel and Schwartzstein, 2018](#); [Bronnenberg et al., 2015](#)). Our main contribution is to jointly model demand and supply (as in [Atal et al., 2024](#)), considering how information frictions interact with segmentation in equilibrium. Our results also suggest that providing information about contraceptive quality can reduce manufacturer's ability to exert market power.

Third, while a rich literature in health economics has studied demand-side barriers to contraceptive use, ranging from financial constraints ([Willage, 2020](#); [Bailey, 2012](#); [Pace et al., 2016](#); [Snyder et al., 2018](#); [Bailey et al., 2023](#); [Dupas et al., 2024](#)) to cultural and social barriers ([Miller et al., 2023](#); [Bau et al., 2024](#)), this work has focused almost exclusively on insurance markets or used randomized controlled trials in low-income settings. Direct evidence from OTC markets remains scarce despite their prevalence across countries, and the supply-side role of market structure in shaping contraceptive access has received little attention ([Rossin-Slater et al., 2026](#)). We fill this gap by providing the first evidence

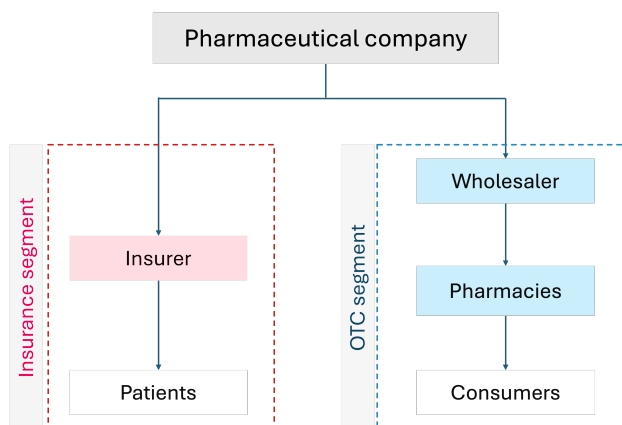
on how segmentation between OTC and insurance markets affects women's access to contraception, with direct implications for the regulation of OTC contraceptive approvals.

2 Setting

2.1 Institutional Background

We study the welfare consequences of market segmentation in the Colombian hormonal contraceptive market, where products are sold through both OTC and insurance segments. The OTC segment accounts for roughly 70% of contraceptive sales by volume, and the insurance segment makes up the remaining 30% (Appendix Figure 1). Contraceptives in Colombia are known as “branded generics,” meaning that products are off-patent, yet are sold under distinct brand names rather than under their chemical designation. Segmentation between OTC and insurance therefore does not reflect a distinction between branded and generic drugs.

FIGURE 1: Pharmaceutical Supply Chain in Colombia



Insurance segment. In the insurance segment, health insurers negotiate wholesale prices with pharmaceutical manufacturers or distributors for contraceptives covered by the national health insurance plan (see Figure 1). The set of contraceptives covered by the health plan is determined by the government on the basis of clinical effectiveness, and is therefore exogenous to individual insurers. Approximately 60 brand-dosage combinations (here-

after, “products”) are covered by the health plan, including first- and second-generation oral contraceptives, implants, intrauterine devices (IUDs), and injections.

First- and second-generation oral contraceptives contain older progestin formulations associated with higher rates of side effects such as weight gain, acne, and irregular bleeding, whereas third- and fourth-generation formulations use newer progestins with milder androgenic side effects (Sitruk-Ware, 2006). Notably, third- and fourth-generation oral contraceptives are not covered by the national health insurance plan. When women obtain a covered contraceptive, insurers pay the negotiated wholesale price to the manufacturer or distributor, but women face zero out-of-pocket costs.

Health insurance coverage in Colombia is nearly universal: roughly 93% of Colombians are enrolled (DANE, 2019), so selection into insurance is limited. All insurers are required by law to cover the same set of contraceptives, so women face the same formulary regardless of which insurer they belong to. The decision of which insurer to enroll with is therefore orthogonal to contraceptive demand, and the relevant selection margin is whether women choose to obtain contraceptives OTC or through their insurer.

Accessing contraceptives through insurance requires a doctor’s prescription, which takes time and effort to obtain. Doctors recommend contraceptives based on the woman’s health status and family planning needs, which supports treating women as the primary decision-makers.

OTC segment. In the OTC segment, pharmaceutical manufacturers negotiate wholesale prices with distributors, who then supply pharmacies. Pharmacies set the retail prices faced by consumers, though strong competition among pharmacies suggests that pricing power rests primarily with manufacturers rather than retailers.⁶ Women purchasing OTC have access to the full set of government-approved contraceptives (including those offered through insurance), nearly 150 products.

Some OTC products have historically been considerably expensive. For example, before 2019, the most expensive fourth-generation oral pill had a retail price equivalent to 15% of

⁶See for example https://www.bloomberglinea.com/latinoamerica/colombia/el-auge-de-los-genericos-la-estrategia-de-ahorro-que-transforma-el-mercado-farmaceutico/?reader_id=READER_ID&url=SOURCE_URL&_RANDOM/

the monthly minimum wage. Although a doctor's prescription is technically required for pharmacy purchases, enforcement is minimal, making it a *de facto* OTC market.⁷ Women purchasing OTC therefore face high prices but few hassle costs, while women sourcing them from insurance face zero out-of-pocket prices but substantial hassle costs.

Based on the structure of the market, women must first select a market segment and then a product within that segment. This choice is inherently complex. It requires trading off prices and hassle costs for segment choice and then learning about contraceptive quality through experimentation for product choice, but experimentation is costly because switching products can result in unintended pregnancy. Hence, women often face information frictions about quality when choosing their contraceptive.

2.2 Price Ceiling Policy

To facilitate contraceptive choice and reduce out-of-pocket costs for women purchasing OTC, the Colombian government imposed wholesale price ceilings on 29 contraceptives in January 2019. Half of the targeted products were third- and fourth-generation oral contraceptives—commonly marketed as higher quality—whose prices prior to the policy were nearly three times higher than in countries such as Canada and Norway ([Ministerio de Salud y Protección Social, 2019](#)). The regulation was subsequently expanded in July 2019 and May 2020 to cover 13 additional products, bringing the total number of regulated contraceptives to 42.⁸

The contraceptive price ceilings were implemented under a broader pharmaceutical price regulation framework introduced in 2013, which targeted products based on two criteria. First, products had to belong to highly concentrated markets—with either fewer than three competitors or a Herfindahl-Hirschman Index exceeding 2,500—and have the highest market share in their market.⁹ Second, their retail prices had to exceed an international reference price, defined as the 25th percentile of average prices for the same active

⁷An audit study in Bogotá found that only 1% of pharmacy staff requested a prescription when selling contraceptive pills ([Andia et al., 2022](#)).

⁸Additional details on regulated products are provided in Circulares 07/2018, 09/2019, and 10/2020 from the Ministry of Health available at [MinSalud, 2025](#).

⁹Markets were defined as the fifth level of the World Health Organization's Anatomical Therapeutic Chemical (ATC) classification (ATC5).

ingredient across a set of comparator countries.¹⁰

3 Data

We use three datasets for our analysis. The first is IQVIA Colombia, which reports monthly retail prices, quantities, national drug codes (NDCs), brand names, manufacturers, and routes of administration for all contraceptives (3-digit ATC G03) sold OTC between 2018 and 2019, covering approximately 90% of Colombian pharmacies. The retail pharmacy market consists of large pharmacy chains with stores in multiple cities, which account for 30% of units sold in the data, and small independent drugstores, which account for the remaining 70%. While pharmacy chains report retail prices directly, independent pharmacies report only wholesale prices. Hence, to approximate retail prices for these observations, we apply a 15% markup following IQVIA's guidance.

The second dataset are the quarterly adverse event reports from the FDA's Adverse Event Reporting System (FAERS). We use these reports—which cover unintended pregnancies, venous thromboembolism, weight gain, acne, bleeding, device expulsion, and other outcomes—to construct a measure of contraceptive quality at the level of active ingredient, route of administration, and quarter. For active ingredients approved in Colombia but not in the U.S., we supplement the FAERS data with Colombia's National Drug and Safety Agency (INVIMA) adverse event reports. Our quality measure is the Proportional Reporting Ratio (PRR) typically used in pharmacovigilance (Evans et al., 2001). The PRR compares the share of a drug's adverse event reports attributable to a given reaction relative to the share across all other drugs. Appendix 1 provides details on the construction of this measure.

The third dataset are individual-level enrollment and health claims records from the Colombian Ministry of Health and Social Protection (known as *Base Única de Afiliados* and *Base de Suficiencia*, respectively). These data cover all outpatient, inpatient, and prescription claims made by nearly 25 million enrollees in Colombia's contributory health

¹⁰Reference countries include Argentina, Brazil, Chile, Ecuador, Mexico, Panama, Peru, Uruguay, Spain, the United States, the United Kingdom, Australia, Canada, France, Norway, Germany, and Portugal.

insurance system between 2018 and 2019. For each claim, we observe the contraceptive prescribed, and the woman's insurer, age, and municipality of residence.

TABLE 1: Summary Statistics by Market Segment

Variable	Contraceptive type	Pre 2019-01		Post 2019-01	
		mean	sd	mean	sd
<i>Panel A. Over-the-counter</i>					
<u>Targeted</u>					
Prices	Conventional pill	11.44	(7.84)	4.53	(3.56)
	Other	3.66	(1.63)	3.25	(1.53)
Sales volumes	Conventional pill	13.34	(18.97)	18.39	(23.42)
	Other	24.52	(20.05)	35.20	(24.25)
<u>Untargeted</u>					
Prices	Conventional pill	5.33	(5.83)	4.52	(4.77)
	Injection	4.47	(2.54)	4.94	(2.59)
	Other	2.77	(1.64)	2.79	(1.64)
Sales volumes	Conventional pill	7.26	(21.90)	8.19	(24.82)
	Injection	44.98	(32.16)	47.49	(41.45)
	Other	21.22	(33.75)	19.24	(33.97)
Total sales volume		1450.50	(87.67)	1654.09	(84.78)
<i>Panel B. Insurance</i>					
<u>Targeted</u>					
Prices	Conventional pill	3.47	(0.93)	2.30	(0.24)
	Other	1.49	(1.17)	1.66	(0.75)
Sales volumes	Conventional pill	1.91	(3.02)	3.11	(4.17)
	Other	1.74	(2.52)	1.72	(2.21)
<u>Untargeted</u>					
Prices	Conventional pill	0.74	(1.12)	0.75	(1.25)
	Injection	0.81	(0.27)	0.81	(0.28)
	Other	0.67	(0.13)	0.65	(0.02)
Sales volumes	Conventional pill	2.89	(5.97)	2.92	(5.94)
	Injection	20.11	(20.16)	16.62	(22.91)
	Other	20.79	(41.01)	23.13	(42.38)
Total sales volume		289.54	(31.85)	260.06	(62.11)

Note: Table shows mean and standard deviation in parentheses of prices and sales volumes by contraceptive type, regulation status, and policy period from January 2018 to December 2019. Panel A shows summary statistics for the OTC segment and Panel B for the insurance segment. In the insurance segment we exclude data for implants that were introduced for the first time after January 2019. Average prices are weighted by sales volumes. Prices are measured in 2018 USD, and sales volumes are in 1,000 units.

We define a product as a brand–dosage combination and aggregate the IQVIA data to the product level. We similarly aggregate individual-level health claims to the product level, restricting attention to claims made by women of reproductive age (15–45). To ensure comparability across contraceptive presentations, we standardize prices and quantities to

a single menstrual cycle. For example, for an NDC corresponding to three blisters of 21 pills, we divide the price by three and multiply the quantity by three. Appendix 1 provides additional details on data cleaning.

Table 1 reports summary statistics for the OTC and insurance segments before and after the first regulatory wave in January 2019. An observation in this table is a product-month. Prior to the regulation, targeted products in the OTC segment had substantially higher prices than untargeted products. For example, the average price of a targeted conventional pill (\$11.44) was more than twice that of an untargeted one (\$5.33). After the policy, prices of targeted products declined to levels comparable to those of untargeted products, and their sales volumes increased. The policy also generated spillover effects, as prices of untargeted products within the same contraceptive type as targeted ones declined after the regulation. Overall, total sales volume in the OTC segment increased 14%. In the insurance segment, wholesale prices for targeted contraceptives also declined after the policy, although the effects were smaller in magnitude. For example, the average wholesale price of a conventional pill declined from \$3.47 to \$2.30. Spillover effects are not apparent in this segment.

4 Reduced-Form Evidence of Market Segmentation

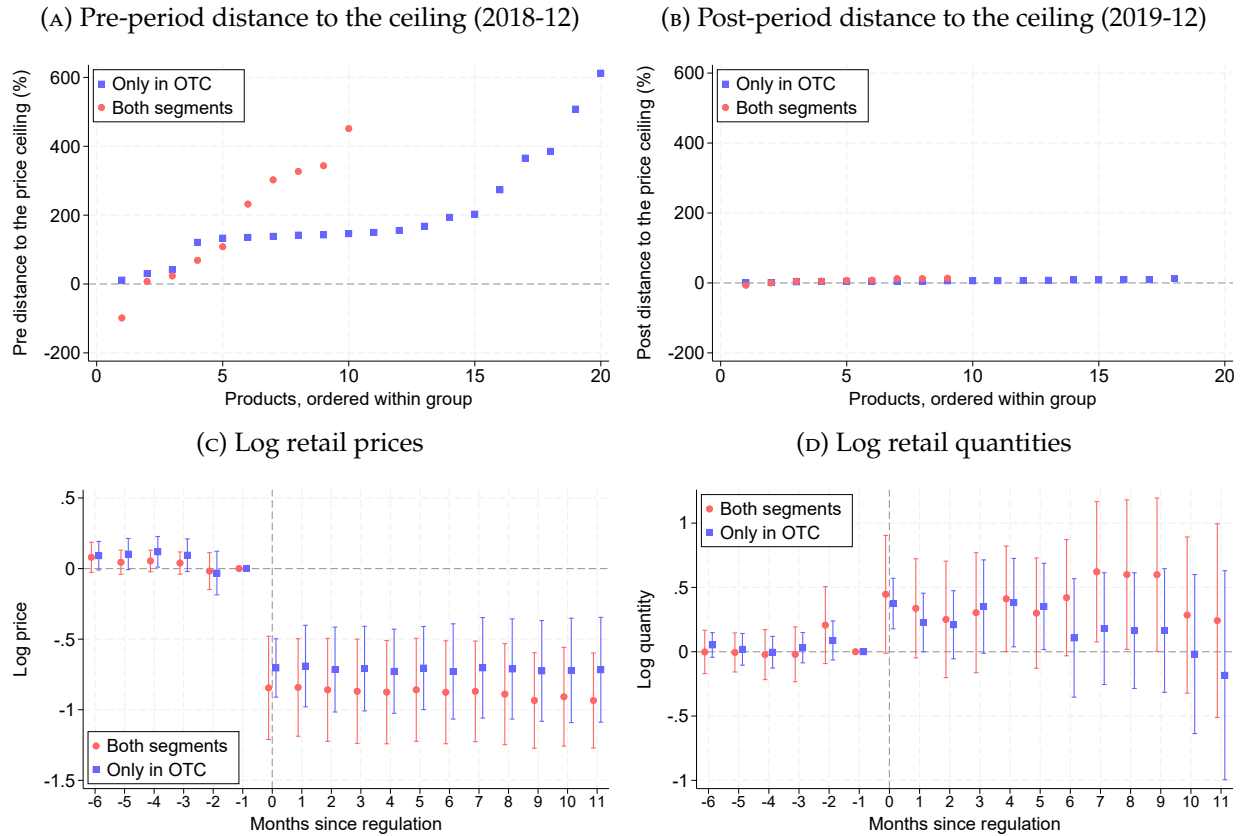
As a first step, we use the price ceiling policy to document the potential equilibrium consequences of market segmentation in reduced form. We estimate an event study on the OTC data, comparing products targeted by the price ceilings against untargeted products:

$$y_{jt} = \beta_{\tau} + \beta_{\tau}^D \cdot \text{Dual}_j + \delta_j + \lambda_{b(j)t} + \gamma_t + \varepsilon_{jt},$$

where y_{jt} is an outcome for product j in month t , τ is the number of months since the regulation wave in January 2019 (normalizing untargeted products to -1), Dual_j indicates products sold simultaneously in both segments, and δ_j and γ_t are product and month fixed effects, respectively. We include fixed effects $\lambda_{b(j)t}$ for the interaction between a product's pre-policy (in $\tau_t = -1$) price decile and calendar month, so that

treatment effects are identified within groups of products priced similarly before the regulation. The coefficients β_τ capture the response of OTC-exclusive products, while β_τ^D measures the differential response of dual-segment products. Because the price ceilings were set as a function of pre-policy prices, $\lambda_{b(j)t}$ prevents differences in nominal ceilings from mechanically generating differences in estimated treatment effects across segments. If segmentation shields OTC-only manufacturers from competition, then β_τ^D should be negative, as OTC-only products start farther from their ceilings. We focus on a balanced panel of products, use a two-way fixed effects estimator, and cluster standard errors at the product level.

FIGURE 2: Heterogeneous Effects of the Price Ceiling Policy in the OTC Segment

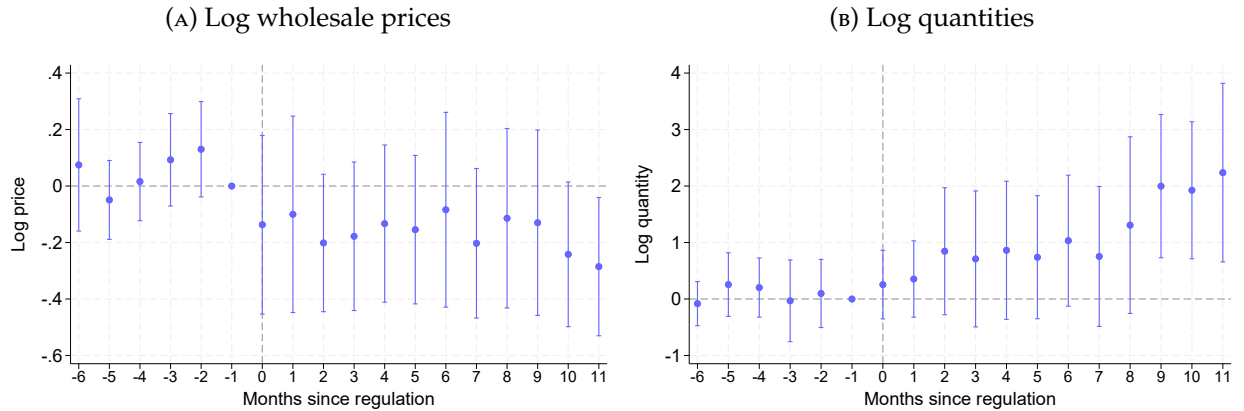


Note: Panel A presents the distance from the retail price to the price ceiling in December 2018 for targeted products. Panel B presents the distance from the retail price to the price ceiling in December 2019 for targeted products. In both panels, products are ordered based on the distance and presented separately for the group offered OTC exclusively (blue) the group offered in both segments (red). Each dot corresponds to one product. Panel C presents event study coefficients and 95% confidence intervals of a TWFE estimator on log retail prices. Panel D reports the analogous estimates for log retail unit sales. An observation in these regressions is a product-month. Estimates in blue report the dynamic treatment effects on OTC-only products. Estimates in red report the dynamic treatment effects on products offered in both segments.

Identification of treatment effects relies on two assumptions: that outcomes for targeted products would have evolved in parallel to untargeted ones absent the regulation (indirectly testable with pre-period estimates), and that the control group is unaffected by the regulation. Because the regulation generated small spillovers onto untargeted products of the same contraceptive type (Table 1), we expect estimates to be slightly conservative.

Figure 2, Panel A starts by showing that OTC-only products on average were priced much higher than the ceiling (203%) compared to dual-segment products (177%). One year after the policy, prices for all products fall to the price ceiling (Panel B). Panel C presents our estimates for β_τ and $\beta_\tau + \beta_\tau^D$. Retail prices for dual-segment products fell by 59% ($= 100 \times (e^\beta - 1)$) in the month after the regulation, compared with 45% for products offered exclusively OTC. OTC-only products adjusted less because segmentation allowed manufacturers to sustain higher prices for these products at baseline. Panel D shows a consistent pattern in quantities though estimates are noisy: dual-segment products, whose prices fell more, also experienced the larger demand increase.

FIGURE 3: Impact of Price Ceilings on Equilibrium Prices and Quantities in Insurance Segment



Note: Figure presents event study coefficients and 95% confidence intervals estimated using a TWFE estimator. Panel A shows the effect of the price ceiling policy on log wholesale prices, and Panel B reports the analogous estimates for log unit sales in the insurance segment. An observation is a contraceptive month.

We next estimate the same event study on claims data from the insurance segment, dropping the Dual_j interaction and the $\lambda_{b(j)t}$ fixed effects, since there are too few insurance-only products to identify these parameters. Figure 3 shows that wholesale prices for targeted products fell 28% (Panel A) and that demand for these products more than doubled after the regulation, particularly toward the end of 2019 (Panel B). The demand

response within the insurance segment is difficult to attribute to prices, since women face zero out-of-pocket costs to begin with. This points to the existence of additional factors shaping contraceptive demand.

5 Survey and Market Evidence on Non-Price Frictions

Information frictions. To understand the forces beyond prices that determine demand through each market segment, we surveyed 296 reproductive-age women in Bogotá in June–August 2026.¹¹ We were interested in measuring women’s awareness of product attributes given that information frictions are a well-known feature of pharmaceutical markets (Crawford and Shum, 2005), and in identifying the main barriers women perceive from sourcing their contraceptives OTC or through insurance.¹²

Respondents were presented with seven commonly marketed contraceptive brands spanning oral pills, injections, and implants. For each brand, we first asked whether they recognized its name and then asked them to identify objectively verifiable characteristics, including route of administration. Appendix Table 2 shows that brand awareness is substantially more common than product knowledge. Although 80.4% of respondents recognize at least one brand, 32.4% cannot correctly identify the route of administration of any recognized brand. Familiarity with a brand name therefore does not imply knowledge of even basic product characteristics, supporting the existence of information frictions (rather than limited consideration set awareness). Knowledge of product attributes also differs markedly by insurance status. Contributors correctly identify an average of 1.07 routes, compared with 0.83 among dependents.

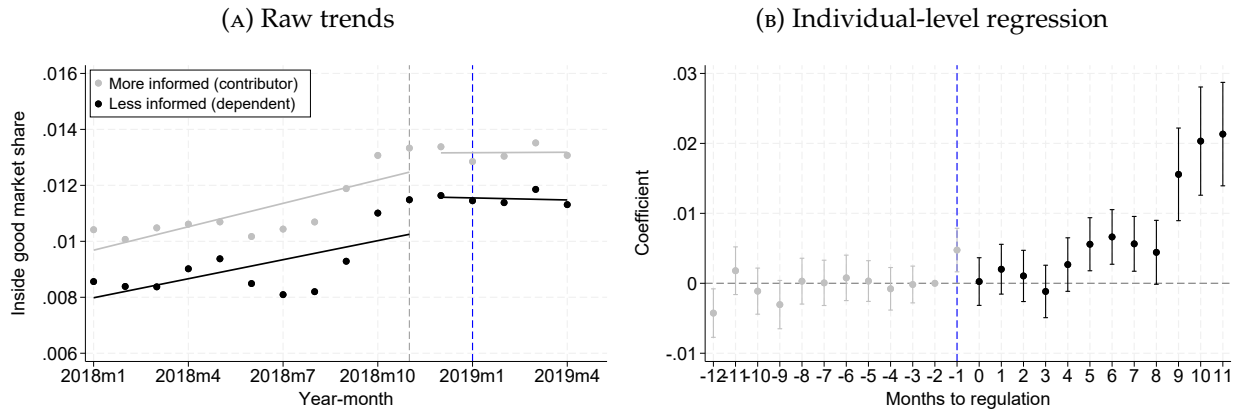
The market data corroborate the survey. Figure 4 compares contraceptive use in the claims data between contributors and dependents, whom we treat as better- and less-informed women, respectively, based on the survey findings. Panel A plots the average

¹¹The original sample consists of 310 respondents. We exclude 14 women who did not report being either contributors or dependents in the contributory health insurance scheme. The resulting analytical sample comprises 296 respondents.

¹²We note that our survey is externally valid: 61.1% obtained their most recent contraceptive from a pharmacy, whereas 13.5% obtained it from an insurer dispensing facility, which are almost the same figures obtained from administrative records such as the Medication Price Information System (SISMED) from the Ministry of Health and Social Protection.

market share of products targeted by the price ceilings, which are considered of higher quality, in each group. Better-informed women exhibit higher baseline use of targeted products, and less-informed women show a stronger post-policy increase, so the gap between the two groups narrows after the regulation. Panel B formalizes these differences in an event study comparing the likelihood of claiming a targeted contraceptive between better- and less-informed women. Less-informed women become 1 to 3 percentage points more likely to claim targeted products after the policy—nearly double the baseline mean. These patterns suggest that women interpreted the policy as a signal of product quality: most targeted products are marketed as having milder side effects, so a regulation singling them out conveyed information that less-informed women did not previously hold. Appendix Figure 2 is consistent with this interpretation, showing a sharp increase in Google searches for the active ingredients targeted by the policy around January 2019.

FIGURE 4: Targeted Product Market Share Among Informed and Uninformed Women



Note: Panel A presents the trend of average targeted product (inside good) market share among contributors and dependents in Colombia's contributory health insurance. Panel B presents coefficients and 95% confidence intervals of an event study comparing targeted product (inside good) market share between dependents and contributors before and after the January 2019 price ceiling policy. The regression includes a treated dummy and month fixed effects, clustering standard errors at the woman level. The dashed blue line represents the month when the price ceiling regulation went into effect.

We explore several competing explanations for the observed increase in demand for targeted products in the insurance data in Appendix 2.1, including changes in insurer product inventories and experimentation, noting that the data are inconsistent with those mechanisms.

Hassle costs. Although information frictions shape demand for contraceptives, they affect women across both market segments to varying degrees, and therefore cannot explain why most sales happen OTC. A more natural explanation is that women face substantial hassle costs in the insurance segment. Obtaining a contraceptive OTC generally requires only a visit to a retail pharmacy, whereas obtaining it through insurance may involve appointments, prior authorizations, or travel to an insurer dispensing facility, which may deter women from claiming medications even when doing so would be financially advantageous.¹³ The survey provides direct evidence of this. Appendix Table 3 shows that 77.3% of respondents have a pharmacy within ten minutes, compared with only 5.2% for an insurer dispensing facility. Conversely, 65.2% report traveling more than thirty minutes to an insurer facility, compared with 5.4% to a pharmacy.

Beyond wait times and travel costs which impact all products equally, hassle costs may also vary systematically across products. Insurers may have stronger incentives to manage utilization of more expensive medications because they are more consequential for profits. If insurers engage in this type of utilization management, women seeking access to more expensive contraceptives would face greater hassle costs than those seeking cheaper alternatives. To examine this, we construct a measure of hassle costs using the health claims data. For women who consume products within a single route of administration, we define the hassle cost as the average number of months she has to wait between consecutive refills conditional on the number of dispensed units. For example, if a woman claims a 3-month pack of pills in January but we see her making her next refill in May, then her wait time is one month. Figure 5 presents a binned scatter plot of this wait time against the wholesale price, after residualizing both variables on route of administration fixed effects to ensure that the correlation is identified from within-route variation rather than across routes.¹⁴ In line with our intuition, contraceptives with higher wholesale prices tend to have longer wait times.

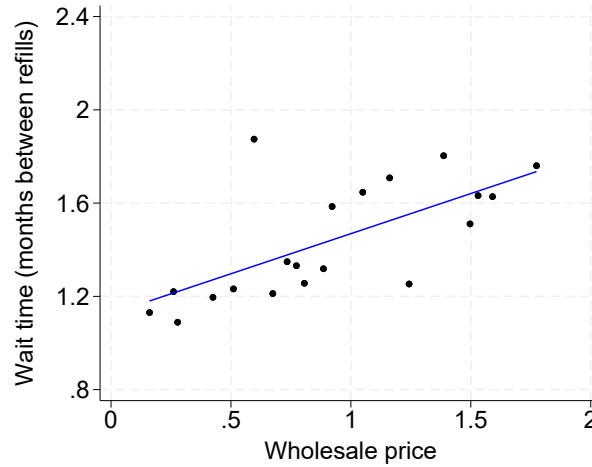
The correlation admits other interpretations as well. For instance, higher-priced con-

¹³In Colombia, long wait times and administrative burdens in medication dispensing are among the most common complaints filed against health insurers (See [El Colombiano, Dec. 21, 2023](#)).

¹⁴This approach explicitly avoids making comparisons between short- and long-acting methods as well as between contraceptives that are typically prescribed for one versus three months.

traceptives may require more complex clinical assessments before being prescribed, or women using more expensive contraceptives may have stronger preferences for specific products and therefore are willing to wait longer to obtain them rather than switching to a cheaper alternative. Although we cannot rule out these explanations, Appendix Table 4 shows that at the product-level, conditional on quality, wholesale price is a significant determinant of product market share even if women do not face these prices directly. Given this evidence, we use wholesale prices as a proxy for the hassle costs women face when accessing contraceptives through insurance in our model.

FIGURE 5: Correlation Between Wait Time and Wholesale Price



Note: Figure presents a binned scatter plot of wait time between refills against wholesale prices in the insurance segment. We start with the individual-level health claims data and use contraceptive claims for women aged 15-50 who consume contraceptives of only one route of administration during the sample period. For each woman we construct the number of months between consecutive prescriptions. Then, we average the refill gap and the wholesale price per product and month. Finally, we construct the binned scatter plot on 20 equally-sized bins of the wholesale price, reporting the average refill gap in the bin.

6 Model

The reduced-form analyses so far yield four main takeaways: (i) market segmentation is consequential for prices and demand by affecting the degree of competition, (ii) contraceptive demand responds strongly to OTC retail prices, (iii) women face information frictions about quality and update beliefs based on quality signals related to the price ceiling regulation, and (iv) women face hassle costs when accessing contraceptives through

insurance. In this section, we develop a model of the Colombian market for hormonal contraceptives that incorporates these takeaways.

6.1 Demand

Contraceptives are supplied through two segments, indexed by $h = \{ins, otc\}$, where *ins* denotes insurance and *otc* denotes over-the-counter. The set of products available to consumers varies across segments, with products indexed by $j = 1, \dots, J_{ht}$, and $|J_{ht}|$ denoting the cardinality of the set. Assume that, conditional on the set of products available in month t through market segment h , women have common prior beliefs about contraceptive j 's quality given by:

$$q_j | J_{ht} \sim \text{Beta}(\mu, \omega),$$

where mean quality is $E[q_j | J_{ht}] = \mu/(\mu + \omega)$.¹⁵ Women receive a signal every month, denoted r_{jt} , about whether contraceptive j is regulated. The product-specific signal depends on the quality of the drug,¹⁶ with

$$r_{jt} \sim \text{Bernoulli}(q_j).$$

Suppose that woman i can be either informed ($v = I$) or uninformed ($v = U$) about product quality. Conditional on the signal and the set of available products, women form Bayesian posterior beliefs about the quality of contraceptive j . For informed women, $v = I$, posterior beliefs coincide with true product quality, $E_{It}[q_j | J_{ht}, r_{jt}] = \theta_j$. For uninformed women, $v = U$, given the distributional assumptions, the mean of the posterior distribution is

$$E_{Ut}[q_j | J_{ht}, r_{jt}] = \sigma E[q_j | J_{ht}] + (1 - \sigma)r_{jt},$$

where $\sigma = (\mu + \omega)/(\mu + \omega + 1)$.

We model contraceptive choice among women who wish to avoid pregnancy using a

¹⁵Our choice of beta-distributed priors follows the data patterns reported in Appendix Figure 4.

¹⁶This assumption is supported by descriptive evidence in Appendix Table 5 showing that higher-quality products are more likely to be regulated.

random coefficients logit model. The indirect utility of woman i from choosing contraceptive j in month t is

$$u_{ijt} = \alpha_i \log(p_{jt}^{\text{eff}}) + \tau \log(p_{jt}^{\text{eff}}) \mathbb{1}\{j \in \text{ins}\} + \gamma E_{v(i)t}[q_j | J_{ht}, r_{jt}] + \rho_i \mathbb{1}\{j \in \text{IUD}\} + \xi_{jt} + \varepsilon_{ijt},$$

where $\alpha_i = \alpha_0 + \alpha_1 \log(a_i)$ and $\rho_i = \rho_0 + \rho_1 \log(a_i)$. The variable $p_{jt}^{\text{eff}} = \{p_{jt}, w_{jt}\}$ is the effective price the woman faces for product j , which is equal to the retail price p_{jt} when the product is supplied OTC and equal to the wholesale price w_{jt} when the product is supplied through insurance. $E_{v(i)t}[q_j | J_{ht}, r_{jt}]$ is the woman's posterior belief about product j 's quality, and $\mathbb{1}\{j \in \text{IUD}\}$ is an indicator for IUDs.

The term ξ_{jt} is the unobserved product characteristic. We decompose this term as $\xi_{jt} = \xi_j + \xi_t + \Delta \xi_{jt}$, where the first component captures preferences for time-invariant contraceptive j 's characteristics, the second captures time-shocks, and the third is a structural demand error. We normalize the utility of not buying contraceptives to zero, so that $u_{i0t} = \varepsilon_{ijt}$, and assume that ε follows a type-I extreme value distribution.

Define the mean utility and the woman-specific component of utility, respectively, as:

$$\begin{aligned} \delta_{jt} &= \alpha_0 \log(p_{jt}^{\text{eff}}) + \tau \log(p_{jt}^{\text{eff}}) \mathbb{1}\{j \in \text{ins}\} + \rho_0 \mathbb{1}\{j \in \text{IUD}\} + \xi_j + \xi_t + \Delta \xi_{jt}, \\ \mu_{ijt} &= \alpha_1 \log(p_{jt}^{\text{eff}}) \cdot \log(a_i) + \gamma E_{v(i)t}[q_j | J_{ht}, r_{jt}] + \rho_1 \mathbb{1}\{j \in \text{IUD}\} \cdot \log(a_i). \end{aligned} \quad (1)$$

Then, the model's predicted market share of contraceptive j is:

$$\hat{s}_{jt} = \int s_{ijt} dP(v, a), \quad (2)$$

where s_{ijt} is the probability that woman i chooses contraceptive j , which is given by:

$$s_{ijt} = \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{|J_t|} \exp(\delta_{kt} + \mu_{ikt})}.$$

6.1.1 Identification

We estimate two types of parameters, one associated with the mean utility and the other with the woman-level component of utility: $\Phi = (\Phi^l, \Phi^{nl})' = ((\alpha_0, \tau, \rho_0), (\alpha_1, \rho_1, \gamma, \sigma))'$.

A. Linear parameters, Φ^l . To identify the linear parameters, we first fix Φ^{nl} and recover the mean utility term δ_{jt} by iterating on a contraction mapping between model-predicted market shares in equation (2) and observed market shares (Berry et al., 1995). Then, we estimate equation (1) and address price endogeneity using as instrument for the effective price of product j the number of rival products within the same ATC code and market segment. The intuition is that a larger number of competitors within the same active ingredient class intensifies price competition, putting downward pressure on the prices of all products in that class. We instrument for the interaction of the effective price with the insurance segment indicator with the analogous interaction with this instrument.

B. Non-linear parameters, Φ^{nl} . We discuss the identification of the nonlinear parameters beginning with beliefs and preferences for quality and then turning to demographic interactions.

B.1. Beliefs and preference for quality. A key identification challenge in our model with information frictions is to separately identify preferences for quality, captured by γ , from imperfect information about quality, captured by σ , since their effects on women's choices are observationally equivalent. To separately identify preferences from information frictions, we exploit the individual-level insurance data to construct two micro-moments that target the difference-in-differences estimator from Figure 4.¹⁷ Intuitively, the fraction of informed women (contributors to the health insurance system) who consume products targeted by the price ceiling policy identifies the preference for quality, while the difference in the fraction of uninformed women (dependents in the health insurance system) who consume targeted products before and after the policy identifies the relative weight

¹⁷This indirect inference approach follows Fu and Gregory (2019) who match regression discontinuity estimates.

women place on information signals. The respective micro-moments are:

$$\text{For } \gamma: \quad g^{\text{qual}}(\Phi) = \frac{1}{T} \sum_{t=1}^T \hat{m}_t^{\text{qual}}(\Phi) - \bar{m}^{\text{qual}} = 0,$$

$$\text{For } \sigma: \quad g^{\text{reg}}(\Phi) = \left(\frac{1}{T_{\text{post}}} \sum_{t \in \text{post}} \hat{m}_t^{\text{reg}}(\Phi) - \frac{1}{T_{\text{pre}}} \sum_{t \in \text{pre}} \hat{m}_t^{\text{reg}}(\Phi) \right) - (\bar{m}_{\text{post}}^{\text{reg}} - \bar{m}_{\text{pre}}^{\text{reg}}) = 0,$$

with:

$$\begin{aligned} \hat{m}_t^{\text{qual}}(\Phi) &= \frac{\frac{1}{R} \sum_i \sum_j \mathbb{1}\{j \in \text{reg}\} \cdot \mathbb{1}\{v_i = I\} \cdot \hat{s}_{ijt}(\Phi)}{\frac{1}{R} \sum_i \sum_j \mathbb{1}\{j \in \text{reg}\} \cdot \hat{s}_{ijt}(\Phi)}, \\ \hat{m}_t^{\text{reg}}(\Phi) &= \frac{\frac{1}{R} \sum_i \sum_j \mathbb{1}\{j \in \text{reg}\} \cdot \mathbb{1}\{v_i = U\} \cdot \hat{s}_{ijt}(\Phi)}{\frac{1}{R} \sum_i \sum_j \hat{s}_{ijt}(\Phi)}, \end{aligned}$$

where $\mathbb{1}\{j \in \text{reg}\}$ is an indicator for whether product j is targeted by the price ceiling policy; R denotes the number of women sampled for estimation, set to 500; and T denotes the number of months. The empirical targets $\bar{m}^{\text{qual}}$, $\bar{m}_{\text{post}}^{\text{reg}}$ and $\bar{m}_{\text{pre}}^{\text{reg}}$ are computed from the claims data.

B.2. Demographic interactions. To identify the parameters capturing individual heterogeneity in preferences, we use another set of micro-moments capturing the consumption patterns by age and route of administration. Let $\hat{s}_{it}^c(\Phi) = \sum_j \mathbb{1}\{\text{route}_j = c\} \hat{s}_{ijt}(\Phi)$, for $c \in \{\text{pills, injection, IUD}\}$, the micro-moments are:

$$g^{a,c}(\Phi) = \frac{1}{T} \sum_{t=1}^T \hat{m}_t^{a,c}(\Phi) - \bar{m}^{a,c} = 0, \quad g^{\tilde{a},w}(\Phi) = \frac{1}{T_{\text{pre}}} \sum_{t \in \text{pre}} \hat{m}_t^{\tilde{a},w}(\Phi) - \bar{m}^{\tilde{a},w} = 0,$$

with

$$\widehat{m}_t^{a,c}(\Phi) = \frac{\frac{1}{R} \sum_i \log(a_i) \cdot \hat{s}_{it}^c(\Phi)}{\frac{1}{R} \sum_i \hat{s}_{it}^c(\Phi)}, \quad \widehat{m}_t^{\tilde{a},w}(\Phi) = \frac{\frac{1}{R} \sum_i \sum_{j \in N_{\text{ins}}} w_{jt} \hat{s}_{ijt}^{\tilde{a}}(\Phi)}{\frac{1}{R} \sum_i \sum_{j \in N_{\text{ins}}} \hat{s}_{ijt}(\Phi)}, \quad \forall \tilde{a} \in \{\leq 35, > 35\}$$

where N_{ins} is the number of product-month observations in the insurance segment and the targets $\overline{m}^{a,c}$ and $\overline{m}^{\tilde{a},w}$ are computed from the health claims data.

Finally, we bring the matrix of exogenous variables from the inner loop that identifies Φ^l , namely $\mathbf{Z}_d$, to the outer loop to form the moment condition:

$$\mathbf{g}^{\text{agg}}(\Phi) = \frac{1}{N} \mathbf{Z}_d' \xi(\Phi) = \mathbf{0}$$

where N is the total number of product-month observations. These moments help identify Φ^{nl} through δ_{jt} .

6.2 Estimation Approach

We use a generalized method of moments (GMM) estimator that combines standard demand-side instruments with micro-moments derived from the price ceiling policy and our individual-level health claims data. Collecting all the moment conditions presented above in the vector $\mathbf{g}(\Phi)$, our GMM estimator solves the following objective function:

$$\widehat{\Phi} = \arg \min_{\Phi} \mathbf{g}(\Phi)' \mathbf{W}_d^{-1} \mathbf{g}(\Phi)$$

The weighting matrix $\mathbf{W}_d$ is block-diagonal, with each block equal to the empirical variance of the corresponding micro-moment and a block equal to $\mathbf{Z}_d' \mathbf{Z}_d$ for the aggregate moment conditions.¹⁸

¹⁸Because the two types of moments are on different scales, creating weighting imbalance, we rescale $\mathbf{Z}_d' \mathbf{Z}_d$ by a normalization factor $\kappa = \frac{\bar{\sigma}_{\text{micro}}^2}{\bar{d}_{\text{agg}}}$ where $\bar{\sigma}_{\text{micro}}^2$ is the mean empirical variance across the micro-moments, and $\bar{d}_{\text{agg}}$ is the mean diagonal element of the aggregate block. The rescaled aggregate block is then $\kappa \cdot \mathbf{Z}_d' \mathbf{Z}_d$.

6.3 Supply

The OTC contraceptive supply chain involves price negotiations between manufacturers and wholesalers and between wholesalers and retail pharmacies (see Figure 1). However, we do not observe these negotiated prices in the data. Because IQVIA reports only retail prices, we model manufacturers in the OTC segment as setting retail prices as if they were vertically integrated with pharmacies. This assumption is reasonable since retail pharmacy markets in Colombia are highly competitive.

In the insurance segment, manufacturers and/or wholesalers sell to health insurers and negotiate prices. Because the OTC segment is manufacturers' largest and most profitable segment, we assume that manufacturers are willing to supply insurers only if they can capture most of the gains from trade. We assume, for simplicity, that manufacturers contract directly with insurers and possess all the bargaining power. We further assume that there is a single representative insurer. Manufacturers make take-it-or-leave-it (TIOLI) offers to the insurer, specifying a per-unit wholesale price. The insurer can accept or reject the offer depending on whether the offer satisfies its participation constraint.¹⁹

Manufacturer f chooses retail prices in the OTC segment and wholesale prices in the insurance segment to maximize total profits given by:

$$\Pi_{ft}(\mathbf{p}^{\text{eff}}) = \sum_{j \in J_{ft, \text{otc}}} (p_{jt} - mc_{jt}) \cdot s_{jt}(\mathbf{p}^{\text{eff}}) M_t + \sum_{k \in J_{ft, \text{ins}}} (w_{kt} - mc_{kt}) \cdot s_{kt}(\mathbf{p}^{\text{eff}}) M_t$$

where $J_{ft,h}$ is the set of products manufactured by firm f in month t and segment h , $\mathbf{p}^{\text{eff}}$ is the vector of effective prices across all firms, mc_{jt} is the marginal cost of product j in month t , s_{jt} is the product's market share, and M_t is the potential market size. We calibrate M_t using contraceptive prevalence estimates from the Demographic and Health Survey, which the World Bank uses to estimate that 83% of reproductive-age women in Colombia use contraception (World Bank, 2025). Hence, the outside option of not using contraceptives has a market share of 17%.²⁰

¹⁹We do not incorporate any incentive compatibility constraints for the insurer in this case because the set of products that are covered through insurance is exogenously determined by the government.

²⁰Our definition of the outside option considers condoms and sterilization as forms of contraception.

In the OTC segment, contraceptive manufacturers compete on retail prices in a Nash–Bertrand fashion. In the insurance segment, manufacturers choose wholesale prices to maximize profits subject to the insurer’s participation constraint. Following [Gowrisankaran et al. \(2015\)](#), we write the insurer’s surplus function as

$$U_t(J_{t,ins}; \varphi) = -(\alpha_a + \tau)^{-1} M_t \underbrace{\int \mathcal{I}_{it,ins} dP_{l(i)}(v, a)}_{\bar{\mathcal{I}}_t} - \varphi_t M_t \sum_{k \in J_{t,ins}} w_{kt} s_{kt}(\mathbf{p}^{\text{eff}}),$$

where $\mathcal{I}_{it,ins}$ is the individual-specific inclusive value from the logit demand, given by

$$\mathcal{I}_{it,ins} = \log \left(\sum_{j \in J_{t,ins}} \exp(\delta_{jt} + \mu_{ijt}) \right).$$

The term $\bar{\mathcal{I}}_t$ denotes the inclusive value aggregated across consumers. We convert it into monetary units by dividing by the absolute value of the price coefficient in the insurance segment, $-(\alpha_a + \tau)^{-1}$, and scale it by market size, M_t . The parameter $\varphi_t > 0$ captures the weight the insurer places on total contraceptive expenditures relative to women’s expected utility. The insurer offers the bundle $J_{t,ins}$ to its enrollees whenever its surplus is nonnegative. We treat this product bundle as exogenous because it is determined by the government.

The Lagrangian of the manufacturer’s problem is as follows:

$$\max_{p_{jt}, w_{kt}} \mathcal{L} = \sum_{j \in J_{ft,otc}} (p_{jt} - mc_{jt}) s_{jt}(\mathbf{p}^{\text{eff}}) M_t + \sum_{k \in J_{ft,ins}} (w_{kt} - mc_{kt}) s_{kt}(\mathbf{p}^{\text{eff}}) M_t + \lambda (U(J_{ins,t}; \varphi) - 0) \quad (3)$$

where λ captures the marginal value of insurer’s profits to the manufacturer. Our equilibrium concept is a Nash equilibrium. Profit maximization involves a set of first-order conditions (FOCs) that after stacking and writing in matrix notation yields:

$$\mathbf{0} = \begin{bmatrix} \mathbf{s}_{otc} \\ \mathbf{s}_{ins} \end{bmatrix} + \begin{bmatrix} \Lambda_{p,otc} & \Lambda_{p,ins} \\ \Lambda_{w,otc} & \Lambda_{w,ins} \end{bmatrix} \begin{bmatrix} \mathbf{p} - \mathbf{mc}_{otc} \\ \mathbf{w} - \mathbf{mc}_{ins} \end{bmatrix} + \lambda \left(\begin{bmatrix} \mathbf{0} \\ \nabla_w \bar{\mathcal{I}} \end{bmatrix} - \varphi \begin{bmatrix} \nabla_p C \\ \nabla_w C \end{bmatrix} \right)$$

Here, Λ is the matrix of demand derivatives with respect to prices multiplied by the ownership matrix, $\nabla_w \bar{I}$ is the derivative of the inclusive value with respect to the wholesale price, and $\nabla_w C$ and $\nabla_p C$ are the derivatives of total contraceptive costs in the insurance segment with respect to the wholesale price and the retail price, respectively. The manufacturer's optimization problem in equation (3) applies to the months before the price ceiling policy. After the policy, the derivatives of OTC contraceptive demand with respect to prices are zero, since the price ceiling is binding.

The FOC of the manufacturer's problem yields an expression for the implied marginal cost of each product:

$$mc_{jt} = p_{jt}^{\text{eff}} + \Delta_{jt}^{-1} s_{jt} - \lambda \cdot \nabla U_t$$

We decompose the marginal cost as $mc_{jt} = x_{jt} \beta^s + \omega_{jt}$, where x_{jt} are observed product characteristics including the product's age, whether it is imported, and the number of cycles it covers; β^s are supply-side parameters to be estimated; and ω_{jt} is the supply-side structural error capturing unobserved cost shifters. Substituting, we obtain the estimating equation:

$$p_{jt}^{\text{eff}} + \Delta_{jt}^{-1} s_{jt} - \lambda \cdot \nabla U_t = x_{jt} \beta^s + \omega_{jt} \quad (4)$$

Additional considerations. We do not model how segmentation arises from the OTC approval process or from the determination of insurance formularies. Segmentation could reflect the selection of manufacturers or products into each segment, or differences in the women each segment serves. We take the first two as given and focus on how women's demand responds to the resulting market structure. Two features of the setting support this. First, we do not model manufacturers' participation in each segment because it does not change over our sample period. Second, the composition of products within each segment is likewise stable because there are no new OTC approvals or changes to the insurance formulary over the sample period. Still, one of our counterfactuals allows manufacturers to discontinue products that become redundant once the market is pooled, which speaks to the role of endogenous product composition.

Identification and Estimation Approach. We estimate the supply parameters $\Theta = (\lambda, \beta^s, \varphi_t)'$ using GMM, taking demand estimates as given (as in [Miller and Weinberg, 2017](#)) and restricting estimation to the pre-policy period from January to December 2018.

The parameter φ_t is identified directly from the insurer's participation constraint. In equilibrium this participation constraint must be binding, otherwise the manufacturer can offer a slightly higher price to the insurer, raising profits and still satisfying the participation constraint. Setting the participation constraint to zero therefore yields a closed-form expression for φ_t :

$$\varphi_t = \frac{\sum_{k \in J_{t,ins}} w_{kt} s_{kt}(\mathbf{p}^{\text{eff}})}{\bar{I}_t}$$

The linear parameters β^s are identified from equation (4) under the assumption that the product characteristics in x_{jt} are exogenous to unobserved cost shifters.

Finally, identification of λ requires an instrumental variable approach. This Lagrange multiplier is potentially endogenous because it reflects the manufacturer's optimal response to cost conditions that are observed by the manufacturer but unobserved to the econometrician. To address this, we instrument λ with the [Gandhi and Houde \(2020\)](#) differentiation instruments constructed from the active ingredient composition of rival products. Specifically, we use the sum of squared differences in grams of progestin and estrogen between product j and all other products in the same market. These instruments are valid for two reasons. First, the hormonal composition of a contraceptive is determined by pharmacological requirements that are orthogonal to a manufacturer's cost shocks, satisfying the exclusion restriction. Second, the hormonal composition of rival products directly determines the degree of substitutability that product j faces, which in turn affects product j 's prices and thus the value of the insurer's participation constraint.

We implement these instruments in our GMM procedure by constructing the moment condition:

$$\mathbf{g}^{\text{supply}}(\Theta) = \frac{1}{N_{\text{pre}}} \mathbf{Z}'_s \omega(\Theta) = \mathbf{0}$$

where the matrix $\mathbf{Z}_s$ includes the exogenous product characteristics x_{jt} and our instru-

ments for λ , and N_{pre} is the number of product-month observations in the pre-policy period. The GMM estimator solves the following objective function:

$$\hat{\Theta} = \arg \min_{\Theta} \mathbf{g}(\Theta)' \mathbf{W}_s^{-1} \mathbf{g}(\Theta)$$

where $\mathbf{W}_s = \mathbf{Z}_s' \mathbf{Z}_s$. Finally, to ensure that the Kuhn-Tucker condition $\lambda \geq 0$ is satisfied, we reparametrize $\lambda = \exp(\tilde{\lambda})$ and estimate $\tilde{\lambda}$ directly. Appendix 3 provides the estimation details.

7 Results

Table 2 reports our estimation results. Women experience negative marginal utility from both retail prices and wholesale prices, but the disutility from wholesale prices in the insurance segment is substantially larger, suggesting prices and hassle costs deter contraceptive take-up. Price sensitivity declines with age, consistent with older women having greater financial stability and a more established contraceptive routine. Although the preference for quality is positive and economically meaningful, it falls below conventional levels of statistical significance. However, we find evidence of substantial information frictions: women place 84% weight on their prior beliefs about product quality and only 16% on new quality signals, indicating that most women remain poorly informed about contraceptive quality in the absence of a policy shock. Finally, the relative appeal of IUDs increases with age.

The second panel of Table 2 reports the supply-side estimates. Marginal costs are higher for imported products than for domestically produced ones, in line with exchange rate exposure, though this effect is imprecisely estimated. Products covering more menstrual cycles have lower marginal costs per cycle. Imposing the insurer's participation constraint to recover φ implies that insurers place a weight of 6% on total contraceptive costs and the remaining 94% on women's surplus.²¹ The estimated Lagrange multiplier λ is small but statistically significant, allowing us to reject pure Bertrand pricing in the insurance

²¹We recover the explicit weight on contraceptive costs by solving for ψ from $\varphi = \frac{\psi}{1-\psi}$.

segment. Appendix Table 6 shows that our model closely matches the empirical micro-moments from the claims data. The fit is somewhat weaker for the quality micro-moment, because part of the variation in quality across products is absorbed by the demand fixed effects, leaving less residual variation for the model to match.

TABLE 2: Model Estimation Results

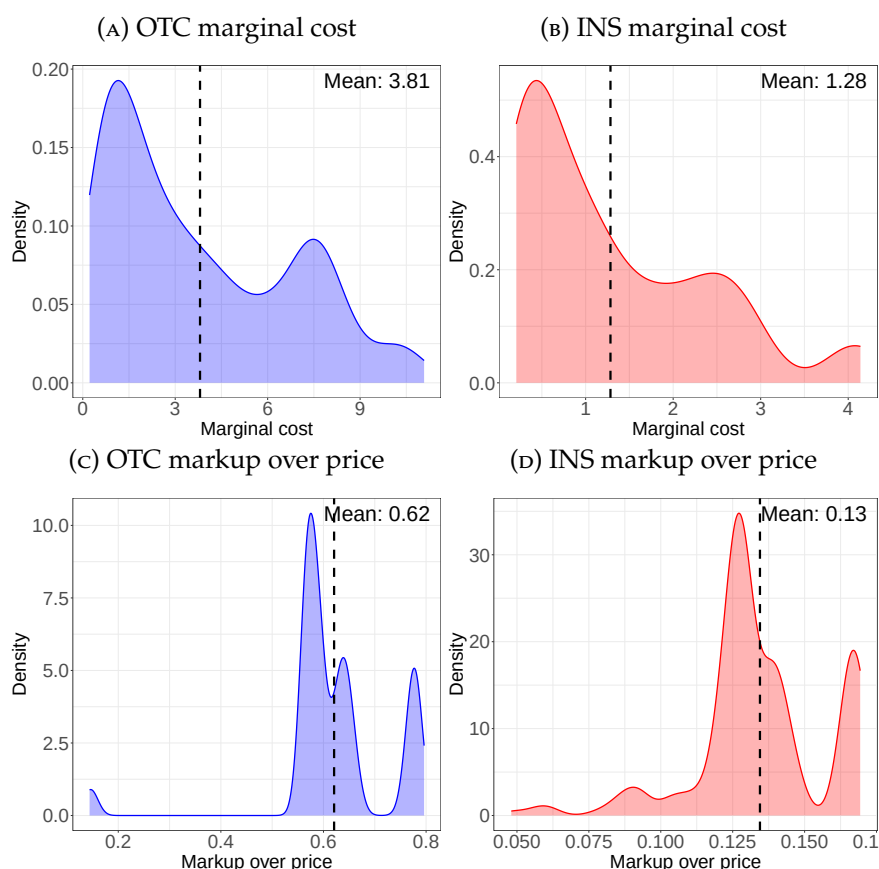
Variable	coef	se
<i>Demand parameters</i>		
Log effective price	-2.159	(0.195)
Log effective price $\times$ Log (age/10)	0.393	(0.079)
Log effective price $\times$ INS dummy	-6.442	(0.107)
IUD dummy $\times$ Log (age/10)	5.793	(0.142)
Quality preference (γ)	3.327	(2.228)
Weight on prior (σ)	0.841	(0.111)
<i>Supply parameters</i>		
Number of cycles	-0.038	(0.024)
Imported product	0.173	(0.332)
Product age	0.064	(0.104)
Shadow price of insurance (λ)	0.0002	(2.2e-8)
Weight on insurer costs (φ)	0.057	[0.048]
<i>Model metrics</i>		
Median own-price elasticity	-1.772	[2.773]
Demand GMM Objective	0.707	
Supply GMM Objective	25.369	
<i>Demand FEs</i>		
ATC-Route-Segment-Lab	Yes	
Month FE	Yes	
N	1975	
<i>Supply FEs</i>		
ATC-Route-Segment-Lab	Yes	
Month FE	Yes	
N	971	

Note: Table presents the demand and supply model estimates. In the demand model we use as instruments for the log effective price and its interaction with the insurance segment dummy the number of competing products in the active ingredient and month and its interaction with the insurance segment dummy. We recover the weight insurer's place on contraceptive costs by setting the participation constraint to zero. The instrument for the shadow price of insurance is average distance between the focal product and its competitor's grams of estrogen and progestin. The regression for the mean utility and the marginal cost include ATC-Route-Segment-Laboratory fixed effects and month fixed effects. The demand model is estimated on data from January 2018 to December 2019. The supply model is estimated on data from January to December 2018. Standard errors are reported in parenthesis and standard deviations in brackets.

Model metrics. Appendix Table 7 reports average own- and cross-price elasticities in the OTC segment during the pre-policy period by route of administration. Own-price

elasticities exceed one in absolute value across all routes and fall within the ballpark of those derived from experimental variation in out-of-pocket contraceptive prices in the U.S. (Bailey et al., 2025). Demand for conventional pills is more elastic than for IUDs and injections, and IUD users are more likely to switch to injections than to conventional pills when prices rise. Appendix Table 8 reports hassle cost elasticities for the insurance segment, which are much larger in magnitude than price elasticities in the OTC segment. Substitution patterns in the insurance segment also broadly mirror those OTC.

FIGURE 6: Distribution of Marginal Costs and Markups



Note: Figure presents the distribution of marginal costs and markups separately by market segment. The supply model estimation uses data from January to December 2018. The dashed line represents the mean of the distribution.

Figure 6 presents the implied distributions of marginal costs and markups across both segments. Marginal costs are bimodal in both segments because of the costs of the two most common active ingredients, levonorgestrel and medroxyprogesterone. The average marginal cost in the OTC segment equals \$3.81, reflecting the combined costs of production

and distribution under our assumption that manufacturers are vertically integrated with pharmacies; in the insurance segment it equals \$1.28, reflecting production costs alone.

Retail markups also have a bimodal distribution with the average equal to 62%. Because marginal costs in this segment include the costs of distribution and pharmacy retail, this markup reflects the supply chain margin in its last link rather than the manufacturer's margin alone. In the insurance segment, where marginal costs capture production costs only, estimated wholesale markups are on average 13%. These two estimates imply that if the manufacturer sold directly OTC, it would earn a markup of nearly 5 times the marginal cost of production.²²

8 Equilibrium Impacts of Market Segmentation

Counterfactual setup. We now use our model estimates to evaluate the equilibrium effects of market segmentation. Later in this section, we extend our analysis to include impacts on fertility. We consider two main counterfactuals, an OTC-only market and an insurance-only market, and report results under three progressively more flexible assumptions. In the DS scenario, we allow demand and supply to adjust while holding product variety fixed. In the DSI scenario, we additionally remove information frictions. In the VDS scenario, we allow product variety to adjust as well. We describe the implementation of each counterfactual below.

In the OTC-only DS counterfactual, we set the insurance segment indicator to zero in the demand function, hence hassle costs have no impact on demand, which puts upward pressure on equilibrium prices. On the supply side, manufacturers price all products according to Nash-Bertrand competition. Two opposing forces shape equilibrium prices in this counterfactual. Removing the insurer's participation constraint allows manufacturers to solve an unconstrained pricing problem, which tends to raise prices. At the same time, manufacturers must offer all insurance-segment products through the OTC segment, intensifying competition across products and putting a downward pressure on prices.

²²This estimate is computed as the difference between the average OTC retail price and the average production cost in the insurance segment, divided by the average production cost in the insurance segment.

The net effect is therefore ambiguous a priori.

In the DSI counterfactual, we additionally remove information frictions by assuming all women are fully informed. The equilibrium price effects of better information depend on the direction of the quality bias. If women systematically underestimate the quality of higher-priced products, then removing information frictions shifts demand toward these products, putting upward pressure on their prices. Conversely, if information frictions disproportionately affect lower-quality products, removing frictions would shift demand away from these products, putting downward pressure on their prices. The net effect therefore depends on the distribution of quality misperceptions across products, which we recover from the structural estimates.

In the VDS counterfactual, we allow product variety to adjust by assuming that manufacturers discontinue products that become redundant when the two segments are pooled. Specifically, if a manufacturer offers the same product in both segments, we assume they retain only the OTC version. Several forces shape equilibrium outcomes in this counterfactual. Eliminating redundant products reduces the number of inside goods, softening within-market competition and putting upward pressure on prices. However, the net effect on demand is ambiguous. If the discontinued products are disproportionately lower-quality, their removal improves the average quality of the remaining choice set, potentially increasing overall contraceptive demand and supporting higher equilibrium prices. Conversely, if discontinued products are disproportionately higher-quality, women would divert to the outside option, reducing demand for inside goods and putting downward pressure on prices.

The insurance-only counterfactual follows an analogous procedure. On the demand side, we set the insurance segment indicator to one, so hassle costs impact demand for all products, putting downward pressure on equilibrium prices. On the supply side, manufacturers price all products to satisfy the insurer's participation constraint, and we allow φ to adjust endogenously from setting this constraint equal to zero. Unlike the OTC-only counterfactual, the two forces shaping equilibrium prices work in the same direction here—higher demand elasticity and the binding participation constraint on all products both push prices down—though the magnitude of the price response is expected to be

modest, since the shadow price of the participation constraint λ is small relative to other parameters in the model.

In the version of the insurance-only counterfactual where product variety adjusts, we assume manufacturers discontinue the OTC version of any product offered in both segments at baseline, retaining only the insurance version. The equilibrium effects on prices and demand in this counterfactual follow a similar logic as the corresponding OTC-only counterfactual and are therefore ambiguous.

Equilibrium. Our estimates describe the observed equilibrium, which raises the question of how far the counterfactuals extrapolate from it. Two features of the model suggest the counterfactuals are safe from this Lucas critique. First, our estimate of λ is small relative to the other supply-side parameters, so imposing or removing the insurer's participation constraint has small impacts in the equilibrium. Second, in the insurance-only counterfactual we adjust φ endogenously so that the participation constraint holds with equality, ensuring the counterfactual is well-defined. We do not prove existence or uniqueness of the counterfactual price equilibrium, but verify numerically that the system of first-order conditions converges to the same fixed point from a range of starting values, as is standard in the literature.

Results. To ensure comparability across scenarios, we report counterfactual results separately for products that are offered OTC at baseline in Table 3 and for products that are offered through insurance at baseline in Table 4. By reporting results in this way we avoid comparisons of retail and wholesale prices. Table 3 shows that moving to an OTC-only market in the DS scenario, reduces the average price from \$5.80 to \$5.39, a decline of roughly 7%, reflecting the intensified competition across a larger set of products. The price reduction is uneven across routes of administration: conventional pill prices fall from \$8.56 to \$7.86, and injection prices decline from \$4.46 to \$4.26, while IUD prices rise from \$2.22 to \$2.39. Incorporating full information to the OTC-only market in column DSI further reduces average equilibrium prices to \$5.25, suggesting that manufacturers exploit these frictions to sustain higher markups and that women's prior beliefs about

quality tend to be biased upward.

These price reductions, however, depend on the assumption that product variety is held fixed. Once we allow variety to adjust by removing redundant products and retaining their OTC versions, average equilibrium prices rise slightly to \$5.90, roughly 2% above the observed scenario. The reduction in the number of competing products weakens within-market competition sufficiently to offset the gains from eliminating segmentation.

TABLE 3: Counterfactual Retail Prices in Baseline OTC Segment

Route	Observed (1)	VDS (2)	DS (3)	DSI (4)
All types	5.803	5.900	5.393	5.254
Conventional Pill	8.564	8.698	7.863	7.727
Emergency Pill	3.066	3.126	2.972	2.907
IUD	2.219	2.206	2.392	2.386
Implant	1.203	1.192	1.187	1.186
Injection	4.464	4.500	4.259	4.119

Note: Table presents the demand-weighted average retail price conditional on products offered OTC at baseline. We report several scenarios: the observed scenario (column 1); the OTC-only counterfactual where product variety, demand, and supply adjust (column 2); the OTC-only counterfactual where demand and supply adjust but product variety is fixed (column 3); and the OTC-only counterfactual where demand and supply adjust and we remove information frictions (column 4). All counterfactual simulations use information from the pre-policy period.

TABLE 4: Counterfactual Wholesale Prices in Baseline INS Segment

Route	Observed (1)	VDS (2)	DS (3)	DSI (4)
All types	0.848	0.867	0.888	0.889
Conventional Pill	1.134	1.227	1.410	1.414
Emergency Pill	0.799	0.784	0.728	0.726
IUD	1.407	1.372	1.353	1.353
Implant	0.671	0.658	0.655	0.655
Injection	0.809	0.799	0.862	0.861

Note: Table presents the demand-weighted average wholesale price conditional on products offered INS at baseline. We report several scenarios: the observed scenario (column 1); the INS-only counterfactual where product variety, demand, and supply adjust (column 2); the INS-only counterfactual where demand and supply adjust but product variety is fixed (column 3); and the INS-only counterfactual where demand and supply adjust and we remove information frictions (column 4). All counterfactual simulations use information from the pre-policy period.

In Table 4, moving to an insurance-only market in the DS scenario leaves overall wholesale prices nearly unchanged relative to the observed scenario, but patterns are starkly different across routes of administration. Prices for conventional pills increase 24% while prices for IUDs decrease 4%. The overall small price effect reflects a key feature

of the insurance pricing environment: under our specification of TIOLI offers to the insurer, there are no pricing externalities across products or manufacturers, so changes in the product set do not affect the competitive environment for products already offered through insurance.

Removing information frictions in the DSI scenario has an equally modest effect, with the average wholesale price increasing slightly from \$0.848 to \$0.850. Nevertheless, analogous to the OTC-only counterfactual, when we allow product variety to adjust by keeping the insurance version of products that offered in both segments at baseline, we find that average wholesale prices rise.

Appendix Figure 5 plots the counterfactual price trends over the pre-policy period. The patterns are stable across months, confirming that the average effects reported in Tables 3 and 4 are not driven by seasonal variation in contraceptive demand. Appendix Tables 9 and 10 report diversion ratios to the outside option in each counterfactual and the observed scenario. Consistent with the previous results, these diversion ratios increase under the VDS scenario but decrease under the DS scenario relative to baseline.

TABLE 5: Counterfactual Demand (in Millions of Women per Year)

Route	Observed	OTC-only			INS-only		
		VDS	DS	DSI	VDS	DS	DSI
All routes	1.722	1.692	1.859	1.707	1.929	2.003	1.986
Conventional Pill	0.661	0.765	1.426	1.334	0.479	0.168	0.163
Emergency Pill	0.308	0.343	0.173	0.152	1.274	0.268	0.261
IUD	0.040	0.039	0.010	0.008	0.005	0.003	0.002
Implant	0.100	0.004	0.001	0.001	0.112	0.025	0.024
Injection	0.612	0.540	0.248	0.212	0.060	1.540	1.536

Note: Table presents the total demand measured in millions of women per year in the observed scenario and the OTC-only and INS-only counterfactuals. Under each counterfactual, we report several scenarios: one where product variety, demand and supply adjust (VDS); one where demand and supply adjust but product variety is fixed (DS); and one where demand and supply adjust and we remove information frictions (DSI). All counterfactual simulations use information from the pre-policy period.

Table 5 reports counterfactual contraceptive demand in millions of women per year. Moving to an OTC-only market in the DS scenario increases total demand from 1.722 to 1.859 million women per year. The demand response is heavily concentrated in conventional pills, whose take-up rises from 0.661 to 1.426 million women, while demand for emergency pills declines substantially from 0.308 to 0.173 million women, suggesting

these products are disproportionately purchased by women who would otherwise use insurance. Incorporating full information (DSI) leaves total demand nearly unchanged relative to the observed scenario, but reallocates demand away from conventional pills and toward injections, consistent with women updating their beliefs in favor of longer-acting methods. When we additionally allow product variety to adjust (VDS), total demand decreases to 1.692 million women, as the removal of redundant products worsens the average quality of the choice set (see Appendix Table 11).

The DS scenario in the insurance-only market increases total demand to 2.003 million women per year, driven almost entirely by a two-fold increase in injection demand as women substitute away from conventional pills whose prices rise in this counterfactual. Assuming women are fully informed about product quality (DSI) leaves aggregate demand nearly unchanged relative to the DS scenario but still meaningfully higher than the observed equilibrium, which suggests that information frictions dampen contraceptive demand. Finally, when product variety adjusts (VDS), total demand equals 1.929 million women per year, which is higher than the observed equilibrium because remaining products in the choice have higher quality than baseline (see Appendix Table 11).

TABLE 6: Counterfactual Consumer Surplus (in Dollars per Woman)

Route	Observed	OTC-only			INS-only		
		VDS	DS	DSI	VDS	DS	DSI
All	0.807	0.926	1.388	1.096	0.397	0.641	0.565
Informed	0.829	0.965	1.403	1.472	0.414	0.643	0.647
Uninformed	0.774	0.867	1.364	0.532	0.371	0.638	0.442
Age 15-19	0.709	0.762	1.182	0.477	0.382	0.649	0.481
Age 20-35	0.798	0.917	1.373	1.141	0.400	0.642	0.579
Age 36+	0.920	1.095	1.613	1.410	0.397	0.631	0.575

Note: Table presents women's surplus per capita measured in dollars per woman in the observed scenario and the OTC-only and INS-only counterfactuals. Under each counterfactual, we report several scenarios: one where product variety, demand and supply adjust (VDS); one where demand and supply adjust but product variety is fixed (DS); and one where demand and supply adjust and we remove information frictions (DSI). All counterfactual simulations use information from the pre-policy period.

Table 6 reports counterfactual consumer surplus in dollars per woman. To convert expected utility to dollar terms, we use the price coefficient α_i in the OTC-only counterfactual and $\alpha_i + \tau$ in the insurance-only counterfactual. In the OTC-only counterfactual, the DS scenario raises average consumer surplus from \$0.807 to \$1.388 per woman, a

72% increase. The gains are broadly shared but somewhat larger for uninformed women (\$1.364, a 76% increase) than for informed ones (\$1.403, a 69% increase). Consumer surplus also increases monotonically with age. Full information (DSI) raises surplus further for informed women to \$1.472 but reduces it for uninformed women to \$0.532, as they reallocate their demand toward higher-priced products. Hence, among uninformed women, the price effect dominates the quality gains. When product variety adjusts (VDS), average surplus falls to \$0.926 relative to the DS scenario, as the reduction in product variety partially offsets the gains from lower prices, though surplus remains 15% higher than in the observed scenario.

In the insurance-only market, the DS simulation reduces average consumer surplus by 21% per woman relative to the observed scenario, because of the impact of hassle costs. The welfare loss is more pronounced for informed women, consistent with their stronger preferences for higher-quality products that now face hassle costs. Imposing full information (DSI) reduces surplus further to \$0.565. Lastly, when product variety adjusts (VDS), surplus continues to fall to \$0.397.

Taken together, our results show that market segmentation dampens contraceptive demand. Pooling the two segments, whether through an OTC-only or insurance-only market, would substantially increase contraceptive uptake. Yet, the welfare implications of eliminating segmentation depend on assumptions about product variety: the gains from pooling markets can be substantially larger when manufacturers are required to maintain the full range of products rather than discontinuing those that become redundant. This suggests that maintaining product variety is as important as limiting price discrimination across segments for determining the welfare effects of market segmentation.

Impacts on fertility. The demand and welfare effects documented above have direct implications for fertility outcomes. To translate our counterfactual results into fertility effects, we draw on estimates from [Ennis and Serna \(2026\)](#), who exploit a shift-share design that compares municipalities with higher versus lower pre-policy exposure to the price ceilings—measured by the share of women consuming targeted products before the regulation—to identify the causal effect of contraceptive prices on total fertility rates (TFR).

We apply their estimated elasticities to the predicted changes in contraceptive prices under each counterfactual scenario to compute the implied effects on fertility.

Table 7 presents the results. Moving to an OTC-only market with fixed product variety (DS) reduces the TFR by 0.713 births per woman on average. This corresponds to an aggregate reduction of 1041 births, with teen births (ages 15–19) falling by 202. Effects are amplified when information frictions are additionally removed (DSI), since the TFR declines by 0.955 and total births by 1394. In contrast, the insurance-only market raises the TFR by 0.468 and increases total births by 634 under fixed product variety (DS). When product variety adjusts in either counterfactual (VDS), the effects on fertility are small and in the opposite direction. These results suggest that the design of contraceptive market regulation has meaningful consequences for fertility outcomes, particularly among teenage women.

TABLE 7: Counterfactual Impacts on Fertility

Counterfactual	Total fertility rate	Total births	Teen births (15–19)
<i>Panel A. OTC-only</i>			
VDS	0.168	245.4	47.5
DS	-0.713	-1041.4	-201.5
DSI	-0.955	-1394.6	-269.8
<i>Panel B. INS-only</i>			
VDS	0.219	319.4	61.8
DS	0.468	683.6	132.2
DSI	0.488	712.3	137.8

Note: Table presents the change in total fertility rate (column 1), total number of births (column 2) and teen births (column 3) implied by the counterfactual average price changes. Treatment effects of percentage changes in prices on fertility outcomes are obtained from [Ennis and Serna \(2026\)](#).

9 Conclusion

This paper studies how market segmentation between over-the-counter (OTC) and insurance channels shapes access to pharmaceuticals with an application to hormonal contraceptives. We use data from Colombia where women can access contraceptives OTC at retail prices with zero hassle costs or through their health insurer at zero prices but with substantial hassle costs. First, we use a price ceiling policy introduced by the govern-

ment in 2019 and a survey instrument to learn about the determinants of contraceptive demand. We show that contraceptive demand is significantly price-sensitive, that women face information frictions about product quality, and that women face hassle costs when sourcing contraceptives through insurance.

Next, to simulate the implications of removing market segmentation, we develop and estimate a structural model that takes into account the determinants of demand. Results show that market segmentation hurts contraceptive take-up and these effects are exacerbated when women lack accurate information about product quality. The reason is that market segmentation permits manufacturers to price discriminate and charge higher prices in both segments. Moving to an OTC-only market raises contraceptive demand by 8% and consumer surplus by 72% while reducing total fertility rates by 0.7 births per woman. Moving to an insurance-only market raises demand by 16% but reduces consumer surplus by 21%, as the hassle costs of interacting with the healthcare system outweigh the benefits of increased product competition. These effects are sensitive to assumptions about product variety: when manufacturers are required to maintain the full range of products rather than discontinuing ones that become redundant after pooling the market, the welfare gains from eliminating segmentation are substantially larger. These findings have direct implications for OTC expansion in other drug classes and countries where pharmaceuticals are also offered through health insurance.

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Online Appendix

Manuscript Title: Segmentation in Pharmaceutical Markets: Evidence from Contraceptives

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Appendix 1 Details on Data Construction

1.1 IQVIA and INVIMA data

The IQVIA dataset reports monthly prices and quantities of contraceptives purchased OTC in Colombia between 2009 and 2023. We have information on all medications categorized under ATC codes G03A, G03C, and G03F.

An observation in the raw dataset is a combination of the national drug code (NDC), month, and pharmacy type. Pharmacy type indicates whether the report originates from a chain or stand-alone pharmacy. A pharmacy is classified as chained if it operates more than five establishments under the same corporate name. Independent pharmacies account for around 70% of OTC contraceptive sales in Colombia. The NDC uniquely identifies a drug and its commercial presentation. For example, a single oral contraceptive brand might have two presentations: one with a single blister containing 21 pills and another with two blisters or 42 pills.

Our data cleaning procedure proceeds as follows:

- *Sample selection from INVIMA.* We begin with the universe of products registered in INVIMA. We exclude NDCs corresponding to hormone replacement therapy (post-menopause) and treatments for amenorrhea. The resulting sample focuses on contraceptive methods, including conventional oral contraceptives, emergency pills, injections, implants, intrauterine devices (IUDs), and patches. The INVIMA dataset contains 1,285 NDCs corresponding to 324 products, where a product is defined as a commercial brand.
- *Imputation of NDCs in IQVIA data.* Due to inconsistencies in pack codes across datasets, we impute INVIMA NDC identifiers to IQVIA pack codes. This process

yields 227 matched NDCs out of 246 pack codes (corresponding to 190 products, including hormone replacement therapy). Unmatched observations account for less than 1.5% of total sales.

- *Merging INVIMA and IQVIA data.* We merge the INVIMA and IQVIA datasets using the matched identifiers, resulting in 183 NDCs corresponding to 162 products after excluding hormone replacement therapy.
- *Price normalization.* We deflate nominal prices using the Colombian Consumer Price Index (base: December 2018 = 100) and convert them to U.S. dollars using the 2018 average exchange rate (2,952 COP/USD).
- *Retail price adjustment.* As prices reported for independent pharmacies correspond to wholesale prices, we apply a 15% retail markup following IQVIA guidelines.
- *Standardization of units.* To ensure comparability across products with different packaging and dosage forms, we standardize prices and quantities to a single menstrual cycle.
- *Aggregation.* We aggregate the data to the product-month level.

After cleaning outliers, the final dataset comprises 156 unique products observed monthly from January 2009 to December 2023, yielding 16,551 observations in an unbalanced panel.

1.2 Enrollment and Health Claims Data

1.2.1 Product-level data

To construct the product-month panel for the insurance segment for the months between January 2018 and December 2019 we use individual-level health claims records. Below is the list of cleaning steps we use to construct this data:

Sample restrictions. We restrict the sample as follows:

- We retain only claims made by women of reproductive age (15–45).
- We retain only women who are always enrolled with insurers in the contributory scheme.
- We restrict attention to outpatient prescription claims for hormonal contraceptives, identified using ATC codes and medication names from INVIMA.
- We drop claims with missing or implausible wholesale prices (zero or negative).
- We determine the number of presentations per refill as the median number of months between refills per woman. For median refills over 3 months, we constrain to 3 months.
- We define a product as a brand-dosage combination. We standardize quantities and prices to a single menstrual cycle to ensure comparability across presentations. For products dispensed in multi-cycle presentations, we divide the price by the number of cycles and multiply the quantity accordingly.

Price cleaning. Wholesale prices in the health claims reflect negotiated contracts between insurers and providers and vary across insurers, providers, and contract types in ways unrelated to the underlying price of the contraceptive. Contract types include fee-for-service, capitation, and fee-for-package. We proceed in two steps to clean these prices:

1. We residualize observed log wholesale prices per claim on contract type–product, insurer–provider–product, and insurer–provider–month fixed effects to remove contract-driven price variation.
2. We define the “cleaned price” as the exponential of the residual of the previous regression, plus the constant, plus the fee-for-service dummy per product.
3. We winsorize the cleaned price at the 2nd and 98th percentiles to remove outliers, and define the final price as the median of the winsorized residual at the product-month level.

Market share construction. We aggregate cleaned claims to the product-month level, computing total cycles dispensed and the final price for each product-month cell.

1.2.2 Individual-level data

For descriptive analyses where we use the individual level data we follow the cleaning steps below.

Sample restrictions. We apply the following restrictions sequentially:

- We retain only women of reproductive age (15–50).
- We retain only women who are always enrolled with insurers in the contributory scheme.
- We retain only women who are continuously enrolled throughout 2018–2019, dropping those who enter or exit mid-sample.
- We drop women who switch insurers or municipalities of residence during the sample period, to ensure stable identification of hassle costs and avoid confounding from insurer or location changes.

Multiple contraceptive claims per month. To deal with the fact that there may be multiple prescriptions for contraceptives in the same day, we prioritize those with greater effectiveness or clinical recommendation following the table below, which was produced by ChatGPT.

Panel construction. We expand the annual enrollment data to a monthly panel, generating one observation per woman per month. We then merge in contraceptive claims, other health claims (outpatient, inpatient, and prescription), and income data from 2017 payroll records.

APPENDIX TABLE 1: Suggested Ranking of Hormonal Contraceptives (High to Low Priority)

Rank	ATC Code	Reasoning
1	G03AD01	Implant – long-acting, high effectiveness
2	G03AC03	Depo-Provera – 3-month injectable
3	G03AC06	2-month injectable
4	G03AA14	Drospirenone progestin-only pill – newer, more selective
5	G03AA07	Desogestrel progestin-only pill – modern, widely used mini-pill
6	G03AA17	Nomegestrol progestin-only pill – selective, less common
7	G03AA12	Lynestrenol progestin-only pill – older generation
8	G03FA11	Combined oral contraceptive with Desogestrel – third-generation progestin
9	G03FA01	Combined oral contraceptive with Levonorgestrel – second-generation progestin

Carryforward of contraceptive use. Because contraceptives differ in their duration of protection, we carry forward contraceptive claims to fill months without a new prescription:

- For three-month injections, we carry forward the claim for up to two additional months.
- For monthly products (pills and one-month injections), we carry forward the claim for one additional month.

1.3 Construction of the Contraceptive Quality Measure

We construct a measure of contraceptive quality at the level of active ingredient, route of administration, and quarter using adverse event reports from the FDA Adverse Event Reporting System (FAERS) from 2014 to 2019. We describe the construction procedure below.

Data assembly. We download quarterly FAERS files covering patient demographics, drug, outcome, reaction, therapy, indication, and report source tables for each quarter from 2014Q1 to 2019Q4. We retain only cases in which the reported indication includes contraception-related terms (e.g., contraception, birth control, oral contraceptive, emergency contraception). For each case, we retain drugs listed as the primary or secondary suspect ($\text{role_cod} \in \{PS, SS\}$) and merge in the associated reaction and outcome records.

Severity classification. We classify adverse events into two categories:

- *High-severity events*: outcomes recorded in the FAERS outcomes table, including death, hospitalization, disability, and life-threatening events.
- *Low-severity events*: reactions recorded in the FAERS reactions table, including side effects such as weight gain, acne, mood changes, and irregular bleeding.

We collapse these counts to the active ingredient-route-year-quarter level, after standardizing drug names to active ingredients using a hand-coded crosswalk that maps brand names and generic names to their active ingredients and routes of administration.

Quality measure. We construct a quality measure based on the Proportional Reporting Ratio (PRR), a standard pharmacovigilance statistic that compares the reporting rate of adverse events for a given drug to the reporting rate across all other drugs in the same period. For active ingredient k in quarter q , the PRR is defined as:

$$\text{PRR}_{kq} = \frac{(a_{kq} + 0.5)/(a_{kq} + b_{kq} + 0.5)}{(c_{kq} + 0.5)/(c_{kq} + d_{kq} + 0.5)}$$

where a_{kq} is the number of high-severity adverse event reports for active ingredient k in quarter q , b_{kq} is the number of low-severity reports for the same ingredient, $c_{kq} = A_q - a_{kq}$ is the number of high-severity reports for all other active ingredients in the same quarter, and $d_{kq} = B_q - b_{kq}$ is the analogous count for low-severity reports. The 0.5 correction is applied to avoid division by zero for active ingredients with sparse reports. We then define true quality as:

$$\text{Quality}_{kq} = \frac{1}{1 + \text{PRR}_{kq}}$$

so that higher values correspond to lower relative adverse event rates and therefore higher quality. We collapse to the active ingredient-route-quarter level by taking the mean quality across cases, and carry forward values for quarters with missing observations using the most recent available value.

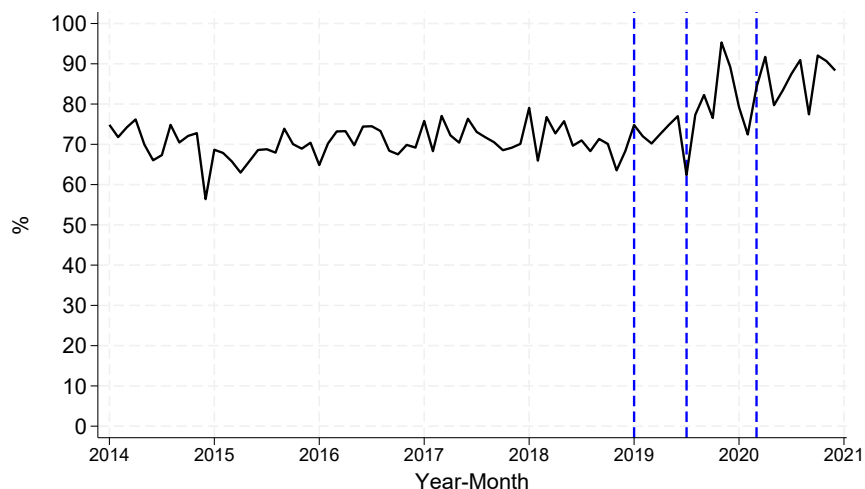
Imputation hierarchy. We construct the quality measure at four levels of aggregation, used hierarchically to impute missing values:

- *Level 1:* active ingredient combination and route of administration (e.g., Levonorgestrel + Ethinylestradiol, oral).
- *Level 2:* progestin component only and route of administration (e.g., Levonorgestrel, oral).
- *Level 3:* active ingredient combination only, pooled across routes.
- *Level 4:* progestin component only, pooled across routes.

For active ingredients approved in Colombia but not reported in FAERS, we supplement with adverse event reports from INVIMA, Colombia's national drug safety agency, using the same PRR methodology (*Level 5*).

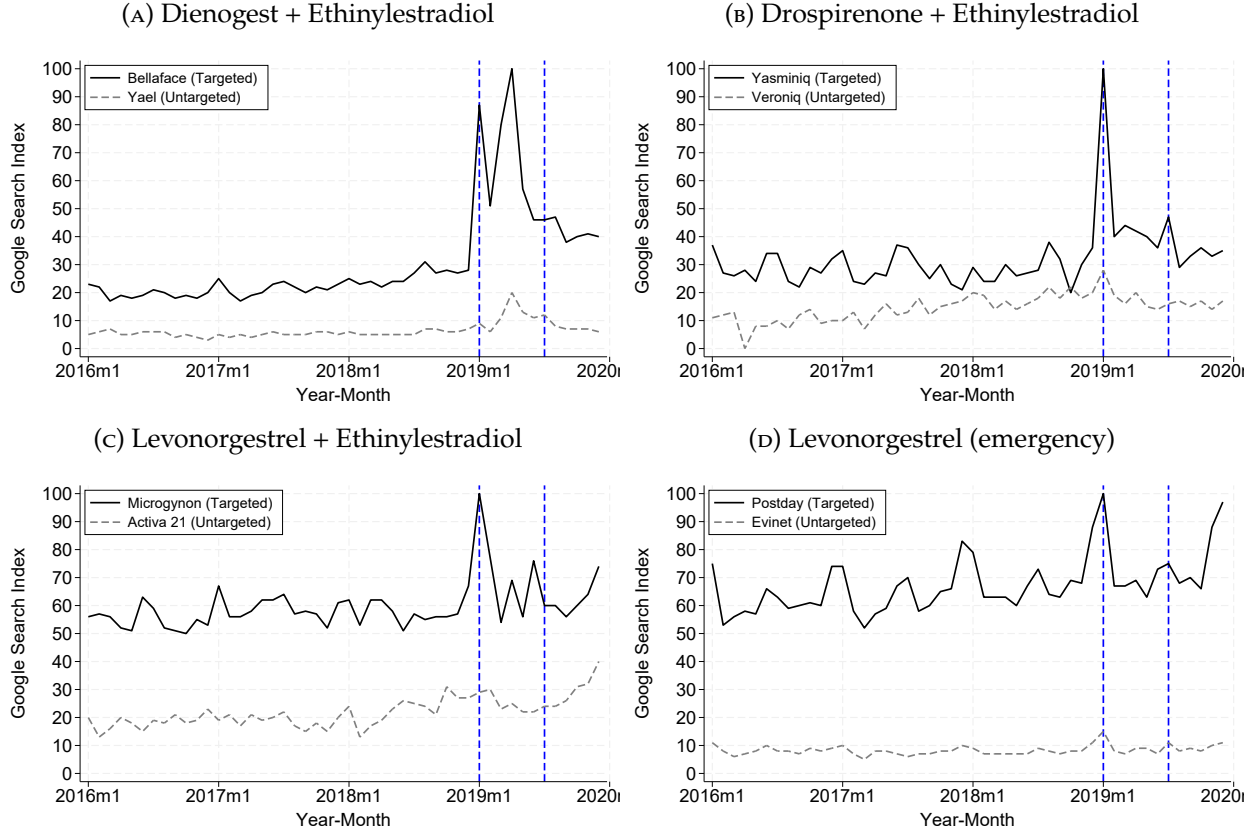
Appendix 2 Additional Descriptive Evidence

APPENDIX FIGURE 1: Share of Total Volume Through the OTC Segment



Note: Figure shows the share of the total volume of contraceptives distributed through the over-the-counter segment. The first blue line marks January 2019, when the first contraceptive price cap regulation (*Circular 07/2018*) came into effect. The second and third lines correspond to July 2019 (*Circular 09/2019*) and March 2020 (*Circular 10/2020*), marking the implementation of two additional contraceptive price cap regulations. Data comes from the Medication and Medical Device Price Information System (SISMED).

APPENDIX FIGURE 2: Google Search Trends on a Subset of Contraceptives



Note: Figure presents the monthly Google Search Index for each pair of contraceptives by active ingredient from 2016-01 to 2019-12. The blue dashed lines correspond to the months when the contraceptive price ceilings were enacted. The light gray, long-dashed line represents untargeted contraceptives, while the solid black line represents targeted products.

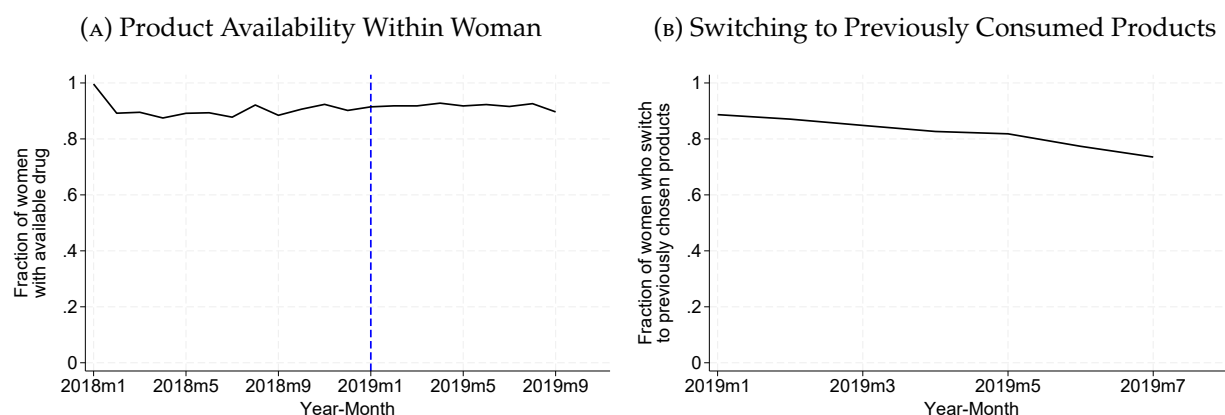
2.1 Competing Explanations in the Insurance Segment

This subsection explores additional competing explanations for the increase in demand for targeted products in the insurance data. First, insurers may have expanded their inventories of targeted products or discontinued untargeted ones, mechanically inducing women to switch after the regulation. To assess this possibility, Appendix Figure 3, Panel A plots, for each month t , the fraction of women whose contraceptive used in month $t - 1$ remained available through their insurer in month t . For approximately 90% of women, their incumbent product continued to be available after the policy, suggesting that forced switching due to product unavailability is an unlikely mechanism. Second, the price ceiling policy may have coincided with a period of high product experimentation among

women. However, Appendix Figure 3, Panel B shows that in January 2019, 90% of women were consuming contraceptives that they had used before 2018, suggesting that product experimentation is also an unlikely mechanism.

The pattern that women return to a previously consumed product after the regulation is also difficult to reconcile with consideration set probabilities. Under a consideration set model, women who were previously unaware of a product would be unlikely to switch to it after the regulation. Yet nearly 90% of switchers choose products they had used before, implying that the relevant barrier was not awareness but rather uncertainty about quality.

APPENDIX FIGURE 3: Evidence of Information Frictions in Contraceptive Use



Note: Panel A presents the fraction of women in month t whose $t - 1$ choice of contraceptive was offered by her insurer-municipality in month t . Panel B presents, among women who switch in month t , the fraction who switches towards products they had consumed before 2019m1. The dashed blue line presents the month when the first price ceiling regulation went into effect.

2.2 Survey Evidence

APPENDIX TABLE 2: Survey Evidence of Contraceptive Knowledge

<i>Panel A. Number of contraceptive routes correctly identified</i>				
	Obs.	Mean	Std. Dev.	Mean as % of max score (%)
Analytical sample	296	0.90	0.77	12.89
Contributor	89	1.07	0.79	15.25
Dependent	207	0.83	0.75	11.87
<i>Panel B. Awareness and route knowledge</i>				
	Obs.	Recognizing at least one brand (%)	Correctly identifying at least one route (%)	Correctly identifying no routes (%)
Analytical sample	296	80.41	67.57	32.43
Contributor	89	87.64	75.28	24.72
Dependent	207	77.29	64.25	35.75

Notes: The table reports awareness of contraceptive brands and knowledge of their routes of administration. Participants were asked about seven contraceptives; consequently, the count variable ranges from zero to seven. A route was classified as correct only when the participant both recognized the contraceptive brand and correctly identified its route of administration. Responses in which the brand was recognized but the route was incorrect or not provided were coded as zero and therefore did not increase the count. In Panel A, *Mean as % of maximum score* is calculated as the mean number of correctly identified routes divided by seven and multiplied by 100. Panel B reports the percentage recognizing at least one brand, correctly identifying at least one route, and correctly identifying no routes. *Contributor* denotes women who finance health insurance coverage for themselves or eligible family members; *Dependent* denotes women covered through another contributor. Respondents who did not report either of these insurance statuses are excluded from the analytical sample.

APPENDIX TABLE 3: Survey Evidence of Hassle Costs

<i>Panel A. Time required to obtain contraceptives</i>		
	Time required to obtain contraceptives	
	Insurance	OTC
Less than one hour	18.18	81.21
More than one hour	81.82	18.79
Observations	33	149
<i>Panel B. Travel time to the nearest location</i>		
	Travel time to the nearest location	
	Insurer delivery point	Retail pharmacy
Less than 10 minutes	5.23	77.29
Between 10 and 30 minutes	29.62	17.29
More than 30 minutes	65.16	5.42
Observations	287	295
<i>Panel C. Location where the most recent contraceptive was obtained</i>		
	Share (%)	Observations
Retail pharmacy or drugstore	61.07	149
Insurer dispensing facility	13.52	33
Other source	25.41	62
Applicable responses	100.00	244

Notes: Entries are percentages of respondents within each column. Panel A reports the time required to obtain the most recent contraceptive, including visits to the pharmacy, waiting in line, or scheduling an appointment with the appropriate physician. The Insurance column includes respondents whose most recent method was obtained at an insurer dispensing facility; the OTC column includes respondents whose most recent method was obtained at a retail pharmacy or drugstore. Panel B reports travel time from the respondent's home or workplace to the nearest insurer delivery point or retail pharmacy. Missing and *Does not know* responses are excluded separately for each location. Panel C reports the location where the respondent obtained her most recent contraceptive among women for whom the question was applicable. OTC denotes over-the-counter.

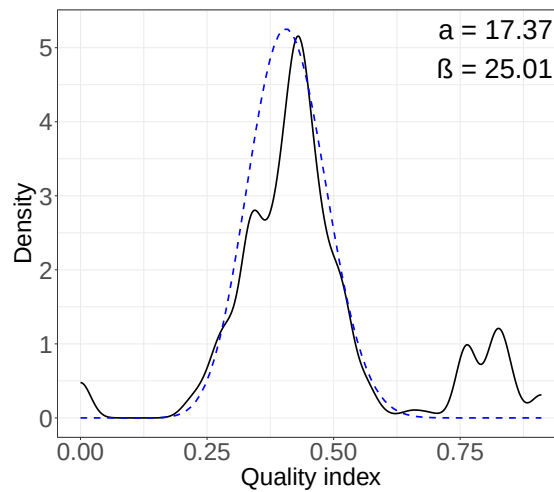
2.3 Support for Model Assumptions

APPENDIX TABLE 4: Correlation between Wholesale Prices and Demand in INS Segment

	Market share	
	(1)	(2)
Log wholesale price	—	-0.004*** (0.0009)
Quality index	0.013 (0.017)	0.003 (0.016)
Route FE	Yes	Yes
Month FE	Yes	Yes
Observations	437	437
R ²	0.689	0.715

Note: Table presents OLS regressions of product market shares in the insurance segment on our measure of quality and the log of wholesale prices. Specifications include route of administration and month fixed effects. Market shares are constructed relative to the size of the potential market. Standard errors in parenthesis clustered at the product level.

APPENDIX FIGURE 4: True Quality Distribution



Note: Figure presents the distribution of true quality in black and an estimated beta distribution in blue. The estimated parameters of the beta distribution are reported in the text box.

APPENDIX TABLE 5: Correlation between Regulation Status and Quality

Variable	$\mathbb{1}\{\text{Regulated}\}$	
	(1)	(2)
Quality index	0.384 (0.071)	0.388 (0.065)
Log price	—	0.113 (0.008)
Route FE	Yes	Yes
Month FE	Yes	Yes
Observations	1,975	1,975
R ²	0.183	0.241

Note: Table presents OLS regressions of an indicator for whether the product is ever regulated on its quality measure in column (1) and additionally on its log effective price in column (2). Both specifications include route of administration and month fixed effects. Robust standard errors in parenthesis.

Appendix 3 Estimation Details

Data preparation. We append the product-month level data from IQVIA and from the insurance segment. Then, we restrict the estimation sample to the pre- and post-policy period from January 2018 to December 2019. We drop all months of products whose maximum inside market share over the sample period falls below 1^{-3} to avoid numerical instability in the contraction mapping. The outside option share is set to 17% of the potential market, consistent with a contraceptive prevalence rate of 83% (World Bank, 2025). We construct two demand-side instruments: the number of competing products within the same month, market segment, and ATC, and its interaction with the insurance segment indicator. The Gandhi and Houde (2020) differentiation instruments with respect to the grams of estrogen and progestin are used as instruments on the supply side.

Demand estimation. We estimate the nonlinear parameters $\Phi^{nl} = (\alpha_1, \rho_1, \gamma, \sigma)$ by minimizing the GMM objective using the BOBYQA algorithm implemented in the `nloptr` package in R (Johnson, 2008). We impose the constraints $\gamma \geq 0$ and $\sigma \in [0, 1]$.

To guard against local minima, we estimate the model from 11 starting values. 10 of these are drawn uniformly from $[-0.1, 0.1]$ for (α_1, ρ_1) , from $[0, 0.1]$ for γ , and from $[0, 0.5]$ for σ . The remaining starting value is set at $(0.4, 4, 4, 0.9)$ from a simple logit regression. We select the solution with the lowest GMM objective value and refine it using tighter convergence tolerances ($\text{ftol_rel} = 1^{-12}$, $\text{xtol_rel} = 1^{-3}$, $\text{xtol_abs} = 1^{-3}$, and maximum 5,000 iterations).

The contraction mapping tolerance is set to 1^{-12} throughout. Conditional on the nonlinear parameter estimates, we recover the linear parameters $\Phi^l = (\alpha_0, \tau, \rho_0)$ via 2SLS as described in equation (1). Estimation is parallelized across 6 cores.

Supply estimation. We estimate the supply-side nonlinear parameter $\tilde{\lambda}$ where $\lambda = \exp(\tilde{\lambda})$ to ensure $\lambda > 0$, by minimizing the supply-side GMM objective using BOBYQA, taking demand estimates as given. We initialize the algorithm at $\tilde{\lambda} = -3$ and use convergence tolerances of $\text{ftol_rel} = \text{ftol_abs} = \text{xtol_rel} = \text{xtol_abs} = 1^{-3}$ with a maximum of 5,000 iterations.

Conditional on $\hat{\lambda}$, we recover the linear supply parameters β^s via OLS on equation (4), and recover φ_t directly from the insurer's binding participation constraint.

Standard errors. We compute standard errors jointly for the demand and supply parameters using the sandwich GMM variance estimator, accounting for the two-step nature of the estimator. The standard error routine takes the estimated nonlinear demand and supply parameters as inputs and computes the full parameter covariance matrix in a single pass.

Model fit. The table below reports the fit of the model on the micro-moments.

APPENDIX TABLE 6: Measures of Model Fit

Moment	Observed	Predicted
$g^{\text{age,pills}}$	0.996	0.975
$g^{\text{age,inject}}$	0.965	0.988
$g^{\text{age,IUD}}$	1.287	1.284
g^{reg}	0.032	0.025
g^{qual}	0.595	0.192
g^{old}	-0.322	-0.256
g^{young}	-0.354	-0.371

Note: Table presents empirical moment from the health claims data and the model-predicted moment.

Appendix 4 Additional Model and Counterfactual Results

APPENDIX TABLE 7: Price Elasticity Matrix by Route of Administration in OTC Segment

Route	Conventional Pills	Emergency Pills	IUD	Implant	Injection
Conventional Pills	-2.080	0.020	0.011	0.002	0.045
Emergency Pills	0.018	-1.755	0.011	0.002	0.045
IUD	0.019	0.020	-1.767	0.002	0.045
Implant	0.018	0.020	0.011	-1.772	0.045
Injection	0.018	0.020	0.011	0.002	-1.730

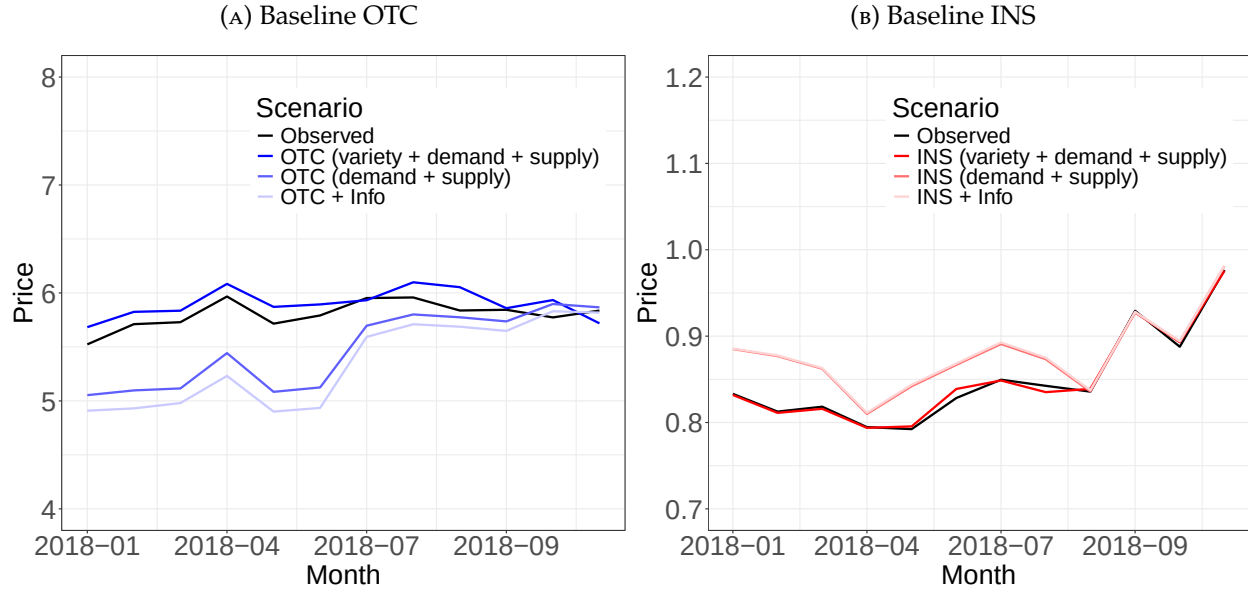
Note: Table presents the price elasticity matrix across routes of administration in the OTC segment during the pre-policy period. To obtain route-level elasticities from the product-level matrix, we aggregate by taking simple averages of own- and cross-price elasticities across all products within each route.

APPENDIX TABLE 8: Price Elasticity Matrix by Route of Administration in INS Segment

Route	Conventional Pills	Emergency Pills	IUD	Implant	Injection
Conventional Pills	-6.826	0.009	0.336	0.012	0.105
Emergency Pills	0.017	-6.839	0.321	0.012	0.105
IUD	0.018	0.009	-6.497	0.012	0.109
Implant	0.018	0.009	0.335	-6.837	0.105
Injection	0.018	0.009	0.344	0.012	-6.739

Note: Table presents the price elasticity matrix across routes of administration in the INS segment during the pre-policy period. To obtain route-level elasticities from the product-level matrix, we aggregate by taking simple averages of own- and cross-price elasticities across all products within each route.

APPENDIX FIGURE 5: Counterfactual Price Trends



Note: Figure presents counterfactual and observed demand-weighted average per month during 2018. Panel A presents trends conditional on products offered OTC at baseline for the OTC-only counterfactual. Panel B presents price trends conditional on products that are offered through insurance at baseline in the INS-only counterfactual. Each panel presents price trends under four scenarios: one where product variety, demand, and supply are allowed to change; one where only demand and supply are allowed to change, and one where demand and supply are allowed to change and information frictions are removed.

APPENDIX TABLE 9: Diversion to Outside Option from Baseline OTC Products

Route	Observed (1)	VDS (2)	DS (3)	DSI (4)
All types	0.141	0.156	0.079	0.156
Conventional Pill	0.141	0.156	0.079	0.155
Emergency Pill	0.140	0.155	0.079	0.158
IUD	0.144	0.159	0.078	0.154
Implant	0.145	0.166	0.083	0.162
Injection	0.140	0.155	0.079	0.156

Note: Table presents the demand-weighted diversion to the outside option conditional on products offered OTC at baseline. We report several scenarios: the observed scenario (column 1); the OTC-only counterfactual where product variety, demand, and supply adjust (column 2); the OTC-only counterfactual where demand and supply adjust but product variety is fixed (column 3); and the OTC-only counterfactual where demand and supply adjust and we remove information frictions (column 4). All counterfactual simulations use information from the pre-policy period.

APPENDIX TABLE 10: Diversion to Outside Option from Baseline INS Products

Route	Observed (1)	VDS (2)	DS (3)	DSI (4)
All types	0.141	0.053	0.007	0.016
Conventional Pill	0.144	0.057	0.007	0.017
Emergency Pill	0.145	0.062	0.007	0.017
IUD	0.145	0.055	0.007	0.016
Implant	0.138	0.050	0.006	0.016
Injection	0.142	0.054	0.007	0.016

Note: Table presents the demand-weighted diversion to the outside option conditional on product offered through INS at baseline. We report several scenarios: the observed scenario (column 1); the INS-only counterfactual where product variety, demand, and supply adjust (column 2); the INS-only counterfactual where demand and supply adjust but product variety is fixed (column 3); and the INS-only counterfactual where demand and supply adjust and we remove information frictions (column 4). All counterfactual simulations use information from the pre-policy period.

APPENDIX TABLE 11: Counterfactual Demand (in Millions of Cycles per Year)

Scenario	Average quality
Observed	0.515
OTC (variety + demand + supply)	0.509
OTC (demand + supply)	0.470
OTC + Info	0.468
INS (variety + demand + supply)	0.605
INS (demand + supply)	0.523
INS + Info	0.522

Note: Table presents the demand-weighted average product quality in the observed scenario and each counterfactual.